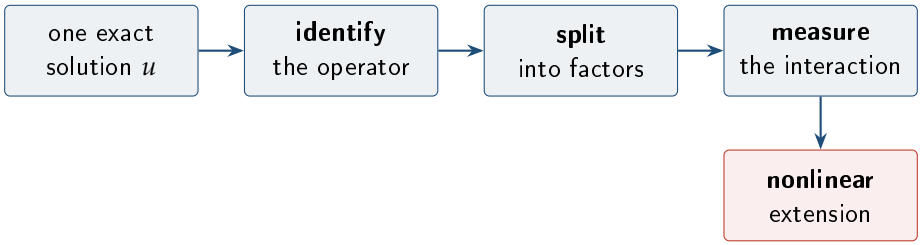
Complete edition · Phases 0–5

Exact Recovery of a Linear Differential Operator from a Single Solution

Identification, factor decomposition, the interaction term, and the nonlinear extension



Companion volume: *Catalogue of proofs, Phases 0–5*. Every theorem below names its proof as "Dim. A*n*" (Part I, main proofs) or "Dim. B*n*" (Part II).

October 2026

Abstract

We are given one exact solution $u(t, x)$ of a differential equation that we do not know. What does u remember about its equation? This edition answers the question completely for the results of Phases 0–5 of the project. The answer has four parts.

Identification. For constant coefficients, u determines its operator P exactly when no partial quotient

$$w_i = \left(\frac{P}{P_i}\right)(\partial)\, u$$

is an exponential-polynomial. No finite amount of data can certify this, and a black-box test is impossible. One extra integer, a bound on the exponential-polynomial part of u, makes it a finite computation.

Decomposition. On a simply connected region, u splits into one component per irreducible *factor* of the symbol. The ambiguity has the exact finite dimension $\delta(P)$. The split per additive *term* of the operator is false. The components keep the Sobolev regularity of u exactly when a conditioning number γ_N is positive.

Interaction. For an evolution equation whose N terms commute, the part of the solution due to the interplay of the terms is

$$E_{\text{int}} = \Big[\prod_{n=1}^{N}(\mathrm{Id} + \Delta_n) - \mathrm{Id} - \sum_{n=1}^{N} \Delta_n\Big] u_0.$$

One value costs $O(N)$ operations, while the full table of cross terms costs $\Theta(2^N)$. The map $u_0 \mapsto E_{\text{int}}$ is not continuous in Sobolev norms and cannot be regularised.

Nonlinear equations. For Burgers, KdV–Burgers and KdV the same questions have sharp answers. The interaction has a t^2 jet and a closed form before the shock, and size $\Theta(\nu)$ and $\nu|\log\nu|$ after it. Identification holds relative to a dictionary, modulo the differential consequences of the equation. For the N-soliton of KdV the identifying polynomial has degree $(N+2)2^{N-1}$.

How to read this edition

Layout. Every chapter opens with a box with three entries. *Starting from* lists the assumptions in force. *Why* says what problem the chapter solves. *Role* says where its output is used. Inside a chapter the order is always the same: motivation, then intuition, then statement, then a worked example, then consequences.

Boxes. The colours are fixed:

- blue boxes are theorems, propositions and lemmas;
- green boxes are definitions;

- violet bars are worked examples;
- amber boxes give the intuition;
- red boxes are warnings and counterexamples.

Proofs. Proofs are not in this volume. Each statement names its proof in the companion *Catalogue of proofs*, for example Dim. A3. Part I of the catalogue contains the 31 load-bearing proofs; Part II contains all the others.

Trust labels. Every statement carries one label. [proved] means a complete proof that passed an independent review. [proved mod. X] means complete except for a cited published theorem X whose source was checked. [proof written, not audited] means a full proof exists but no independent audit has been done. [from cited source] means a published theorem we rely on. [conditional] means the result depends on an unverified lemma. [checked on examples] means it was only checked on examples. [open] means unsolved. [false] means a counterexample is proved. The reviewers were independent AI agents, not human referees. Grey tags such as (C3-T1) are the identifiers in the project ledger `claims/LEDGER.md`.

Prerequisites. Linear algebra (eigenvalues, Jordan form), calculus in two variables, the Fourier transform, and linear ODEs with constant coefficients. Chapter 2 introduces every algebraic notion from scratch: ideals, annihilators, exponential-polynomials and the Chinese remainder theorem.

Scope. This edition covers Phases 0–5. Phases 6–11 (systems, variable coefficients, the analytic encoder, the universal dictionary, convergence, the extended algorithm) are summarised in the separate document *Punto di partenza*. They will be included in the complete edition once Phase 11 is finished.

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CHAPTER 1

Introduction

In this chapter	
Starting from	one function $u(t,x)$, known exactly (no noise) on an open region $\Omega \subset \mathbb{R}^2$, which solves a linear equation $P(\partial_t, \partial_x)u = 0$ with constant coefficients. The operator P is *not* given.
Why	to state precisely what we want to learn from u and to give a map of the answers before any detail.
Role	frames the three questions Q1–Q3. Chapters 4–5 answer Q1, Chapters 6–7 answer Q2, Chapter 8 answers Q3. Chapter 9 extends all three to nonlinear equations and Chapter 10 assembles them into an algorithm.

1.1 The question

Suppose we observe a physical field $u(t,x)$: a temperature, a wave, a concentration. We know u exactly on some region Ω of the (t,x) plane, and we know that it obeys some linear law. We do not know which law. Three questions arise naturally.

Q1. Identification. Can the operator P be reconstructed from u? When is the answer unique?

Q2. Decomposition. If P is built from pieces, can u be split into the parts that each piece generates?

Q3. Interaction. If the pieces are the *terms* of the operator (diffusion, advection, dispersion, ...), which part of u is due to no single term but to their interplay? Can it be computed exactly and cheaply?

The setting is deliberately analytic. There is no statistical model, no learned weights and no noise. The aim is an *exact* procedure, together with a precise statement of what is impossible.

Example 1.1 Three solutions of the heat equation

Take the heat equation $u_t = Du_{xx}$ with $D > 0$. Consider three of its solutions.

- A single Fourier mode:

$$u_1 = e^{-D\kappa^2 t}\cos(\kappa x).$$

- A sum of two modes:

$$u_2 = e^{-Dt}\cos x + e^{-4Dt}\cos 2x.$$

- The heat kernel, for $t > 0$:

$$u_3 = \frac{1}{\sqrt{t}}\, e^{-x^2/(4Dt)}.$$

All three solve the same equation. We will see that only u_3 determines it. The function u_1 also solves $u_{xx} = -\kappa^2 u$ and $u_t = -D\kappa^2 u$, and from u_1 alone we cannot tell which of these laws is "the" law. The difference is that u_1 and u_2 are finite sums of exponentials, while u_3 is not (Chapter 4, Example 4.2).

1.2 Overview of the answers

Figure 1.1 shows the logical chain, read from left to right. Each box is a task. Each arrow is a theorem that allows the next task to be carried out, under its own hypotheses.

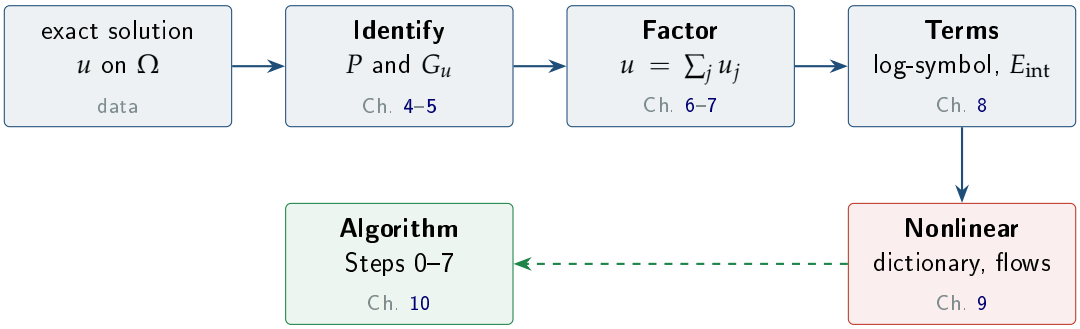


Figure 1.1: Logical structure of the method. The three linear stages (blue) are followed by the nonlinear extension (red). The algorithm (green) chains all of them. A step can only be trusted under the hypotheses of the theorem that precedes it.

Table 1.1 lists the main statements with their status. Every row points to the chapter where the statement is made precise.

1.3 Notation at a glance

Table 1.2 collects the symbols used throughout, with the place where each one is defined. Some letters have different meanings in different chapters; the caption lists them.

1.4 Roadmap

The chapters build on each other in the following order.

- **Chapter 2** is a primer. It explains ideals, annihilators, exponential-polynomials, the Chinese remainder theorem and the companion matrix, starting from integers and one-variable ODEs. A reader who already knows these notions can skip it.

Topic	Result	Status
Identification	$\mathrm{Ann}(u) = (P)$ iff no $w_i = (P/P_i)(\partial)u$ is an exponential-polynomial (Ch. 4)	[proved]
Spectral form	$\mathrm{Ann}(u) = (P)$ iff the point spectrum is empty and $G_u \sim P$ (Ch. 4)	[proved]
Finite data	a jet of order $2d-1$ suffices for exponential-polynomials; no finite certificate otherwise; black box impossible; decidable with a bound (Ch. 5)	[proved]
Decomposition	$\ker P = \sum_j \ker Q_j$ on simply connected Ω, ambiguity $\delta(P)$ (Ch. 6)	[proved]
Terms	splitting per additive term is false (Ch. 6)	[proved]
Regularity	components in H^m iff the sum of kernels is closed ($\gamma_N > 0$) and dense (Ch. 7)	[proved]
Interaction	closed product formula; $O(N)$ per value, $\Theta(2^N)$ per table (Ch. 8)	[proved]
Stability	$u_0 \mapsto E_{\mathrm{int}}$ is not continuous in H^m; no stable regularisation (Ch. 8)	[proved]
Nonlinear	Burgers: jet, closed form before the shock, $\Theta(\nu)$ and $\nu\lvert\log\nu\rvert$ after (Ch. 9)	[proved] / [proved mod. Kružkov]
Dictionary	identification modulo differential consequences; KdV degree $(N+2)2^{N-1}$ (Ch. 9)	[proved] / [proved mod. Novikov]

Table 1.1: Main results, in the order in which they appear.

- **Chapter 3** fixes the algebraic language: symbols, the annihilator ideal, its gcd splitting and the finite-dimension test for exponential-polynomials. It ends with the one-variable prototype.
- **Chapters 4 and 5** answer Q1. The first gives the exact criterion and the intrinsic operator G_u. The second asks what can be decided from finite data.
- **Chapters 6 and 7** answer Q2. The first proves the factor decomposition and shows why splitting by terms fails. The second asks when the pieces keep the regularity of u.
- **Chapter 8** answers Q3: the exact interaction term, its cost, and its instability.
- **Chapter 9** extends the three questions to nonlinear equations.
- **Chapter 10** turns the theorems into an algorithm with two fully worked examples.
- **Chapter 11** states what is impossible, what is open, and how reliable each result is.

Symbol	Meaning	Defined in
(t, x), Ω	time and space; the open region where u is known	Ch. 1
$\partial = (\partial_t, \partial_x)$	partial derivatives	Ch. 3
$\mathcal{R} = \mathbb{C}[\tau, \xi]$	ring of constant-coefficient operators, $\tau \leftrightarrow \partial_t$, $\xi \leftrightarrow \partial_x$	Def. 3.1
P, P_i, e_i	the operator (its symbol), its irreducible factors, their multiplicities	Ch. 3
$Q_j = P_j^{m_j}$	pairwise coprime factors used for decomposition	Ch. 6
$V(P)$	characteristic variety: zeros of the symbol in $\mathbb{C}^2$	Def. 3.1
$\mathrm{Ann}(u)$	the ideal of all operators that annihilate u	Def. 3.2
G_u, J_u	$G_u = \gcd \mathrm{Ann}(u)$ and the cofactor ideal: $\mathrm{Ann}(u) = G_u\, J_u$	Lemma 3.2
EP	exponential-polynomials	Def. 3.3
$\sigma_p(u)$	point spectrum $V(J_u)$	Ch. 4
$\delta(P)$	dimension of the ambiguity of the factor decomposition	Ch. 6
c_F, γ_N	Friedrichs cosine; conditioning number of the sum map	Ch. 7
A_n, $S_n(t)$, Δ_n	terms of an evolution, their flows, increments $S_n - \mathrm{Id}$	Ch. 8
E_{int}	interaction term	Eq. (8.1)
T_n, V, $I(u)$	nonlinear terms; dictionary span; ideal of relations of u	Ch. 9

Table 1.2: Notation guide. Overloaded letters: S is a flow in Chapter 8 and the sum map in Chapter 7; D is d/dx in Section 3.4 and a diffusion constant elsewhere; J denotes ideals.

CHAPTER 2

A primer for readers new to the algebra

In this chapter	
Starting from	only linear algebra, polynomials and linear ODEs with constant coefficients.
Why	the whole theory speaks the language of rings and ideals. Every notion used later is introduced here from a familiar starting point: integers first, then ODEs in one variable, then two variables.
Role	nothing later depends on reading this chapter, but every later chapter uses its vocabulary. It covers ideals and annihilators (§2.1), exponential-polynomials (§2.2), the Chinese remainder theorem (§2.3) and the companion matrix (§2.4).

2.1 Ideals and annihilators, starting from integers

2.1.1 Integers

The multiples of 6 form the set

$$(6) = \{\dots, -12, -6, 0, 6, 12, \dots\}.$$

This set has two properties. The sum of two multiples of 6 is a multiple of 6. A multiple of 6 times any integer is again a multiple of 6. A subset of a ring with these two properties is called an *ideal*.

Now take the ideal generated by 4 and 6, that is, all combinations $4a + 6b$ with integers a, b:

$$(4, 6) = \{4a + 6b : \ a, b \in \mathbb{Z}\}.$$

Since $2 = 6 - 4$ lies in it, and every element is even, we get $(4, 6) = (2)$. The ideal "remembers" only the greatest common divisor of its generators. In $\mathbb{Z}$ every ideal is generated by one number.

2.1.2 Operators in one variable

Replace integers by differential operators with constant coefficients in one variable x, and replace "is a multiple of 6" by "kills the function u".

Definition 2.1 Annihilator (one variable)

For a function $u(x)$, the annihilator of u is the set of all polynomials Q such that the operator $Q(d/dx)$ kills u:

$$\operatorname{Ann}(u) = \Big\{ Q \in \mathbb{C}[\xi] : \; Q\Big(\frac{d}{dx}\Big)u = 0 \Big\}.$$

It is an ideal. If Q kills u and S is any polynomial, then SQ kills u too, because $S(d/dx)$ applied to 0 is 0. Sums of annihilators are annihilators.

Example 2.2 Annihilators of familiar functions

Write $\partial = d/dx$. The operator $\partial - 1$ kills e^x, and so does every multiple of it. The basic cases are:

$$\begin{aligned} \operatorname{Ann}(e^x) &= (\partial - 1), \\ \operatorname{Ann}(\sin x) &= (\partial^2 + 1), \\ \operatorname{Ann}(xe^x) &= \big((\partial - 1)^2\big), \\ \operatorname{Ann}(e^x + e^{2x}) &= \big((\partial - 1)(\partial - 2)\big). \end{aligned}$$

The function e^{x^2} has annihilator $\{0\}$: no nonzero polynomial in ∂ kills it, because each derivative $\partial^k e^{x^2}$ equals a polynomial of degree exactly k times e^{x^2}.

In one variable every ideal has a single generator, exactly as in $\mathbb{Z}$. The generator of $\operatorname{Ann}(u)$ is the *minimal annihilator* of u: the ODE of lowest order that u satisfies.

2.1.3 Two variables: the new feature

With two variables t and x we have two derivatives. An operator is a polynomial in both. Here an ideal may need *several* generators.

Example 2.3 An ideal with two generators

Let $u = e^{2t+3x}$. Then

$$(\partial_t - 2)u = 0 \qquad \text{and} \qquad (\partial_x - 3)u = 0.$$

Every operator of the form $A\,(\partial_t - 2) + B\,(\partial_x - 3)$ kills u, and these are all of them. So

$$\operatorname{Ann}(u) = (\partial_t - 2,\ \partial_x - 3).$$

No single polynomial generates this ideal: a polynomial in the ideal vanishes at the point $(2, 3)$, and a single polynomial vanishes on a whole curve, not at a single point.

This is the source of every subtlety in the theory. A solution of a PDE has an annihilator that may contain a "curve part" (one polynomial) and a "point part" (finitely many points). Lemma 3.2 of the next chapter makes this precise.

2.2 Exponential-polynomials

An *exponential-polynomial* is a finite sum of polynomials times exponentials:

$$w(t,x) = \sum_{j=1}^{r} p_j(t,x)\, e^{\tau_j t + \xi_j x}.$$

These are exactly the solutions of ODEs with constant coefficients, in each variable. They are the "trivial" solutions of the theory: they live on finitely many points of the characteristic variety.

The key property is a finite-dimension test. Differentiate w as often as you like: you always stay inside a finite-dimensional space.

Example 2.4 Which functions are exponential-polynomials?

Function	In EP?	Span of all derivatives
xe^x	yes	span$\{xe^x,\ e^x\}$, dimension 2
$x^2e^{3t} + \cos x$	yes	finite: $\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$
e^{-x^2}	no	$\partial_x^k e^{-x^2} = (\text{degree } k)\cdot e^{-x^2}$: infinite dimension
$1/(1+x^2)$	no	denominators $(1+x^2)^{k+1}$ grow: infinite dimension
$f(x+t)$, f a Gaussian	no	same as the Gaussian

The test "is the span of the derivatives finite-dimensional?" is Lemma 3.3.

2.3 The Chinese remainder theorem

2.3.1 Integers

Solve the system

$$x \equiv 1 \pmod 2, \qquad x \equiv 2 \pmod 3.$$

The unique solution modulo 6 is $x \equiv 5$. In general, if a and b are coprime, a remainder modulo a and a remainder modulo b determine a unique remainder modulo ab. In the language of rings:

$$\mathbb{Z}/(ab) \;\cong\; \mathbb{Z}/(a) \times \mathbb{Z}/(b).$$

2.3.2 ODEs in one variable

Let $p(\partial) = (\partial - 1)(\partial - 2)$. Its solutions are $u = ae^x + be^{2x}$. The two pieces can be extracted by *projector operators*:

$$u_1 = \frac{\partial - 2}{1 - 2}\, u = a\, e^x,$$

$$u_2 = \frac{\partial - 1}{2 - 1}\, u = b\, e^{2x}.$$

Check the first line: $(\partial - 2)(ae^x + be^{2x}) = -ae^x$, and dividing by $1 - 2 = -1$ gives ae^x. The sum is direct:

$$\ker p(\partial) = \ker(\partial - 1) \oplus \ker(\partial - 2).$$

This works because $\partial - 1$ and $\partial - 2$ are coprime: there are polynomials a, b with $a(\partial - 1) + b(\partial - 2) = 1$ (here $a = 1$, $b = -1$).

2.3.3 Two variables: what changes

For the wave operator $P = \tau^2 - \xi^2 = (\tau - \xi)(\tau + \xi)$ the two factors are coprime, but no combination of them equals 1. Indeed

$$(\tau - \xi,\ \tau + \xi) = (\tau,\ \xi),$$

the ideal of the single point $(0, 0)$ where the two lines meet. The quotient

$$\mathcal{R}/(\tau, \xi) \cong \mathbb{C}$$

has dimension 1. This number is the ambiguity of the splitting: the d'Alembert components $f(x + t)$ and $g(x - t)$ are defined only up to a constant moved from one to the other. Theorem 6.3 says that this finite overlap is the *only* failure of the integer picture.

2.4 The companion matrix

A linear ODE of order m becomes a first-order system for the vector of derivatives. For

$$u'' - 3u' + 2u = 0$$

put $X = (u, u')^\top$. Then $X' = CX$ with

$$C = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}.$$

The characteristic polynomial of C is $\lambda^2 - 3\lambda + 2$, the symbol of the ODE, with roots 1 and 2. The Lagrange projectors are

$$\Pi_1 = \frac{C - 2\,\mathrm{Id}}{1 - 2}, \qquad \Pi_2 = \frac{C - \mathrm{Id}}{2 - 1},$$

and the matrix exponential splits into modes:

$$e^{xC} = e^{x}\,\Pi_1 + e^{2x}\,\Pi_2.$$

This is the same formula as the projectors of the previous section, written for matrices. It reappears as the Lagrange formula (6.1) for evolution factors.

Takeaway. Four ideas carry the theory. An *ideal* records all equations a function satisfies. Its *gcd* separates a curve part from finitely many points. *Exponential-polynomials* are the functions with finitely many derivatives up to linear dependence. The *Chinese remainder theorem* splits a solution into pieces, up to a finite overlap.

CHAPTER 3

The algebraic language

In this chapter	
Starting from	constant coefficients; functions or distributions on $\Omega \subset \mathbb{R}^2$ with complex values. The vocabulary of Chapter 2.
Why	three notions carry the whole theory. The *symbol* turns derivatives into polynomials. The *annihilator* collects all equations a function satisfies. *Exponential-polynomials* are the trivial solutions.
Role	foundation for every later chapter. It proves the two lemmas that every identification argument uses (Lemmas 3.2 and 3.3) and ends with the one-variable prototype (Section 3.4).

3.1 Derivatives as polynomials

A linear operator with constant coefficients is a polynomial in the two partial derivatives:

$$P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j\, \partial_x^k, \qquad c_{jk} \in \mathbb{C}.$$

The exponentials are eigenfunctions of every derivative. Write

$$e_{\tau,\xi}(t,x) = e^{\tau t + \xi x}.$$

Then

$$\partial_t\, e_{\tau,\xi} = \tau\, e_{\tau,\xi}, \qquad \partial_x\, e_{\tau,\xi} = \xi\, e_{\tau,\xi},$$

and therefore, for every operator P,

$$P(\partial)\, e_{\tau,\xi} = P(\tau,\xi)\, e_{\tau,\xi}. \tag{3.1}$$

So applying the operator to an exponential is the same as multiplying by the value of a polynomial. This is the reason why all the algebra below is about polynomials in two variables.

Definition 3.1 Operator ring, symbol, characteristic variety

The ring of constant-coefficient operators is

$$\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \;\cong\; \mathbb{C}[\tau,\xi].$$

The polynomial $P(\tau,\xi)$ is the *symbol* of $P(\partial)$. The *characteristic variety* of P is its zero set

$$V(P) = \{(\tau,\xi) \in \mathbb{C}^2 :\; P(\tau,\xi) = 0\}.$$

By (3.1), the exponential $e_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when (τ, ξ) lies on $V(P)$. Fourier modes are the special case $\xi = i\kappa$ with real frequency κ.



Figure 3.1: Real slices of two characteristic varieties. A single exponential solution is a single point of the variety. A rich, non-exponential solution "fills" the whole curve.

3.2 Annihilators and what a solution knows

A single function is usually killed by many operators. For example, e^x is killed by $\partial_x - 1$, by $\partial_x^2 - 1$, by ∂_t, and by every combination of these.

Definition 3.2 Annihilator

The annihilator of u is the set of all operators that kill u:

$$\mathrm{Ann}(u) = \{Q \in \mathcal{R} : \; Q(\partial)u = 0\}.$$

$\mathrm{Ann}(u)$ is an ideal of $\mathcal{R}$: if $Q(\partial)u = 0$ and $S \in \mathcal{R}$, then

$$(SQ)(\partial)u = S(\partial)\big(Q(\partial)u\big) = 0,$$

and the sum of two annihilators is an annihilator. Three classical facts about $\mathbb{C}[\tau, \xi]$ are used throughout.

Three facts about polynomials in two variables

(F1) Unique factorisation. Every nonzero polynomial factors as

$$P = c \prod_i P_i^{e_i},$$

with P_i irreducible and pairwise non-proportional (the "primes"), $e_i \geq 1$ their multiplicities, and c a constant. The factorisation is unique up to constants, so gcd is defined as for integers.

(F2) Bézout. Two polynomials without a common factor have only finitely many common zeros, at most the product of their degrees. Geometrically: two curves without a common component meet in isolated points.

(F3) Nullstellensatz (Hilbert). If a polynomial q vanishes on the common zero set $V(I)$ of an ideal I, then $q^k \in I$ for some $k \geq 1$. In particular, an ideal without common zeros is the whole ring.

All three are classical [1, 2]. They hold over $\mathbb{C}$; the Nullstellensatz fails over $\mathbb{R}$.

The next lemma says that every ideal splits into a "curve part" (one polynomial) and a residue that vanishes only at finitely many points.

Lemma 3.3 Splitting off the gcd (C3-L2) [proved]

Let $I \subset \mathcal{R}$ be a nonzero ideal and $g = \gcd(I)$. Then

$$I = g \cdot J, \qquad J = \{f/g : \ f \in I\},$$

where J is an ideal whose zero set $V(J)$ is finite.

Dim. C3-L2, Catalogue p. 3

Intuition

Divide every element of I by the common factor g. What is left has no common factor, so by Bézout any two "generic" elements of J meet only at finitely many points.

For the annihilator of a function we use a fixed notation:

$$G_u = \gcd \operatorname{Ann}(u), \qquad \operatorname{Ann}(u) = G_u \cdot J_u.$$

G_u is the "genuinely infinite-dimensional" part of the equation of u. J_u records the finitely many exponentials that happen to be present in u.

Example 3.4 Two extreme cases

A single exponential. For $u = e^x$, seen as a function of (t, x):

$$\operatorname{Ann}(u) = (\tau, \ \xi - 1), \qquad G_u = 1, \qquad J_u = \operatorname{Ann}(u).$$

The ideal is that of the single point $(0, 1)$. The function has no infinite part.

A travelling profile. For $u = f(x + t)$ with f a Gaussian:

$$\operatorname{Ann}(u) = (\tau - \xi), \qquad G_u = \tau - \xi, \qquad J_u = \mathcal{R}.$$

The whole ideal is the curve part; there are no isolated points. This is proved in Chapter 4.

3.3 Exponential-polynomials

Definition 3.5 Exponential-polynomials

EP is the set of finite sums

$$w(t,x) = \sum_{j=1}^{r} p_j(t,x)\, e^{\tau_j t + \xi_j x},$$

with polynomials p_j and exponents $(\tau_j, \xi_j) \in \mathbb{C}^2$. We write $\mathrm{EP}(\Omega)$ for their restrictions to Ω.

For a function w, let $\mathcal{R}w$ be the space of all its derivatives and their linear combinations:

$$\mathcal{R}w = \operatorname{span}\{\partial_t^j \partial_x^k w : \ j,k \geq 0\} = \{Q(\partial)w : \ Q \in \mathcal{R}\}.$$

The map $Q \mapsto Q(\partial)w$ goes from $\mathcal{R}$ onto $\mathcal{R}w$ and its kernel is $\operatorname{Ann}(w)$. Hence

$$\mathcal{R}w \ \cong \ \mathcal{R}/\operatorname{Ann}(w).$$

Lemma 3.6 Finite-dimension criterion (C3-L3) [proved]

For a function or distribution w on a connected Ω:

$$w \in \mathrm{EP}(\Omega) \iff \dim_{\mathbb{C}}(\mathcal{R}w) < \infty.$$

In that case $V(\operatorname{Ann} w)$ is the finite set of exponents $\{(\tau_j, \xi_j)\}$ that occur in w.

Dim. C3-L3, Catalogue p. 4

Intuition

($\Rightarrow$) Derivatives of $p\, e^{\tau t + \xi x}$ are polynomials of degree at most $\deg p$ times the same exponential, so they stay in a finite-dimensional space.

($\Leftarrow$) On the finite-dimensional space $\mathcal{R}w$, the derivatives ∂_t and ∂_x act as two commuting matrices. Commuting matrices can be triangularised together. Their joint eigenvalues are the exponents, and their nilpotent parts produce the polynomial prefactors.

Example 3.7 The test on two functions

For $w = xe^x$ the derivatives are $xe^x + e^x$, $xe^x + 2e^x$, and so on: all lie in $\operatorname{span}\{xe^x, e^x\}$, of dimension 2. So $w \in \mathrm{EP}$.

For $w = e^{-x^2}$, the k-th derivative is a polynomial of degree exactly k times e^{-x^2}. These are linearly independent for all k, so $\dim \mathcal{R}w = \infty$ and $w \notin \mathrm{EP}$.

Why it matters. Lemma 3.3 is the computational heart of the method. The question "is w an exponential-polynomial?" is a question of finite-dimensional linear algebra. It is the test used in the identification criterion (Theorem 4.2) and in Step 3 of the algorithm.

3.4 One variable: the ODE prototype

In one variable every object of the later chapters appears with explicit formulas and without the geometry of curves. We consider an ODE $p(D)u = 0$ in the variable x, with $D = d/dx$ in this section only.

Setting

The symbol factors over $\mathbb{C}$ as

$$p(\lambda) = \prod_j (\lambda - \lambda_j)^{m_j},$$

with distinct roots λ_j and multiplicities $m_j \geq 1$. The order is $m = \sum_j m_j$. The solution space is

$$\ker p(D) = \Big\{ \sum_j q_j(x)\, e^{\lambda_j x} : \deg q_j < m_j \Big\}.$$

The form of the solutions comes from one computation. For one root,

$$(D - \lambda)\big(q\, e^{\lambda x}\big) = q'\, e^{\lambda x} :$$

applying $D - \lambda$ lowers the degree of q by one. So $(D - \lambda_j)^{m_j}$ kills $q\, e^{\lambda_j x}$ exactly when $\deg q < m_j$.

Example 3.8 Reading the solution space from the roots

Let $p(\lambda) = (\lambda - 1)^2(\lambda + 2)^3$. The roots are 1 (multiplicity 2) and -2 (multiplicity 3), and the order is 5. A basis of solutions is

$$e^x, \quad xe^x, \quad e^{-2x}, \quad xe^{-2x}, \quad x^2e^{-2x}.$$

3.4.1 Identification by the Wronskian

Proposition 3.9 ODE identification (R4-T1) [proved]

Let $u \in \ker p(D)$. The minimal annihilator of u has degree equal to the number of basis functions actually present in u. It equals p if and only if

$$u,\ u',\ \ldots,\ u^{(m-1)}$$

are linearly independent (nonvanishing Wronskian), that is, iff every root of p appears in u with its full multiplicity.

Dim. R4-T1, Catalogue p. 57

Example 3.10 A missing root

Let $p(\lambda) = (\lambda - 1)(\lambda - 2)$ and $u = 3e^x$. Then $u' = u$, so u and u' are dependent and the minimal annihilator is $\lambda - 1$, of degree 1. The root 2 is missing from u, and u cannot tell p from $\lambda - 1$.

3.4.2 The companion matrix and the spectral form

Write $p(\lambda) = \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_0$ and put $X = (u, u', \ldots, u^{(m-1)})^\top$. Then $X' = CX$ with the companion matrix

$$C = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ -a_0 & -a_1 & \cdots & -a_{m-1} \end{pmatrix}, \qquad X(x) = e^{xC}X(0).$$

The eigenvalues of C are the roots of p. For distinct roots the exponential has the spectral form

$$e^{xC} = \sum_{j=1}^{m} e^{\lambda_j x}\,\Pi_j, \qquad \Pi_j = \prod_{k \neq j} \frac{C - \lambda_k \mathrm{Id}}{\lambda_j - \lambda_k}, \tag{3.2}$$

where the Π_j are the Lagrange projectors onto the eigenspaces.

Proposition 3.11 Projectors of the companion matrix (II-T2, II-T3) [proved]

For an operator monic in τ with pairwise coprime factors, the projectors onto the factor components are explicit rational functions of ξ. Their poles lie exactly where two factor varieties meet. They are polynomial if and only if the varieties are disjoint, that is, iff $(Q_i, Q_j) = \mathcal{R}$ for all $i \neq j$.

Dim. II-T2, Catalogue p. 60

The companion matrix is not normal in general

An early conjecture said that the companion matrix is normal exactly when $p = s^m - c$ with $|c| = 1$. For $m = 2$ this is [false]: the polynomial $s^2 + s - 1$ is a counterexample. The correct normality criterion for $m = 2$ is (II-P5) [proved], and it is formalised in Lean. Consequence: Hilbert-space orthogonality arguments need an inner product adapted to the problem, not the standard one on coefficient vectors.

3.4.3 The jet is a cyclic vector

Proposition 3.12 Cyclic vector (II-C1) [proved]

Let P be monic in τ of degree k. Then $\mathrm{Ann}(u) = (P)$ if and only if the t-jet

$$u,\ \partial_t u,\ \ldots,\ \partial_t^{k-1} u$$

is linearly independent over the ring $\mathbb{C}[\partial_x]$.

Dim. II-C1, Catalogue p. 63

This is the two-variable Wronskian: it replaces "the derivatives are independent" by "the time derivatives are independent with coefficients that are operators in x".

Takeaway. In one variable the polynomial encodes the solution space. Its roots give the exponential rates, the multiplicities give the polynomial prefactors, the companion matrix represents differentiation, and the Lagrange projectors isolate the modes. In two variables the same structure survives. The minimal annihilator becomes the ideal $\mathrm{Ann}(u)$ (Chapter 4) and the Lagrange projectors become Eq. (6.1) (Chapter 6).

CHAPTER 4

Identification: what one solution determines

In this chapter	
Starting from	$\Omega \subset \mathbb{R}^2$ open and *connected*; $u \in \mathcal{D}'(\Omega)$; constant coefficients; an operator P with $P(\partial)u = 0$ is available (any annihilator, not necessarily the "true" one), with irreducible factorisation $P = c\prod_i P_i^{e_i}$.
Why	before splitting a solution into pieces one must know its equation. We ask exactly when u alone pins down P, and what the intrinsic operator of u is when it does not.
Role	first link of the chain in Figure 1.1. Its output is the pair (G_u, J_u), which every later chapter uses, and the test "$\in$ EP", which Step 3 of the algorithm implements.

4.1 The question in algebraic form

Suppose $P(\partial)u = 0$. Then every multiple SP also kills u, so the ideal (P) is contained in $\mathrm{Ann}(u)$:

$$(P) \subseteq \mathrm{Ann}(u).$$

We say that u *identifies* P when the two coincide. In that case every equation satisfied by u is a consequence of P, and P is determined by u up to a constant factor.

Why can the inclusion be strict? Because u may satisfy extra equations that P knows nothing about. The single Fourier mode of the heat equation is the typical case: it also satisfies $u_{xx} = -\kappa^2 u$.

Connectedness of Ω is necessary. On a disconnected region the two pieces can satisfy different equations, and no criterion on u as a whole can see that.

4.2 The criterion

Terminology

- P is *reduced* if all $e_i = 1$ (no repeated factor).
- P is *monic in* τ if its leading coefficient as a polynomial in τ is a nonzero constant. Then $k = \deg_\tau P$ is its order in time, and the *t-jet* of u is the vector $(u, \partial_t u, \dots, \partial_t^{k-1} u)$.

- For each irreducible factor we form the partial quotients

$$w_i = \left(\frac{P}{P_i}\right)(\partial)\, u, \qquad w_i' = \left(\frac{P}{P_i^{e_i}}\right)(\partial)\, u.$$

  In w_i we remove one copy of P_i; in w_i' we remove the whole power.

Theorem 4.1 Identification criterion (C3-T1) [proved]

Let Ω be connected, $u \in \mathcal{D}'(\Omega)$ and $P(\partial)u = 0$.

(a) $\operatorname{Ann}(u) = (P)$ if and only if $w_i \notin \mathrm{EP}(\Omega)$ for every irreducible factor P_i.

(b) $V(\operatorname{Ann} u) = V(P)$ if and only if $w_i' \notin \mathrm{EP}(\Omega)$ for every i.

(c) If P is reduced, (a) and (b) coincide.

(d) If P is monic in τ, the test can be carried out on the t-jet of u (Proposition 3.4.3).

Dim. C3-T1, Catalogue p. 7

Intuition

Strip one irreducible factor off P and look at what is left: w_i lies in the kernel of P_i.

If w_i is a "generic" member of that kernel, not a finite sum of exponentials, then no smaller operator can kill u.

If w_i is an exponential-polynomial, then u contains a finite-dimensional piece. That piece is also killed by an operator of lower order, and u cannot tell the large equation from the small one.



Figure 4.1: The heat curve $\tau = \xi^2$. Left: a solution that is not an exponential-polynomial (the heat kernel) uses the whole curve, and $\operatorname{Ann}(u) = (P)$. Right: a finite sum of exponentials lives on finitely many points. Then $\operatorname{Ann}(u)$ contains polynomials that vanish at those points but are not multiples of P.

Proposition 4.2 Uniqueness of the identified operator

If $\operatorname{Ann}(u) = (P)$, then every operator Q with $Q(\partial)u = 0$ is a multiple $Q = SP$ with $S \in \mathcal{R}$. In particular P is, up to a nonzero constant, the only annihilator of minimal degree.

Dim. C3-T1, Catalogue p. 7

The proof is one line: $Q \in (P)$ means $Q = SP$, and degrees add, so $\deg Q \geq \deg P$ with equality only when S is a constant.

Example 4.3 The heat equation, three cases

Take $P = \tau - D\xi^2$ with $D > 0$. It is irreducible, so $P/P_1 = 1$ and the criterion reads

$$\mathrm{Ann}(u) = (P) \iff u \notin \mathrm{EP}.$$

One Fourier mode. Let $u = e^{-D\kappa^2 t + i\kappa x}$. Then $u_t = -D\kappa^2 u$ and $u_{xx} = -\kappa^2 u$. So u is also killed by $\partial_x - i\kappa$ and by $\partial_t + D\kappa^2$:

$$\mathrm{Ann}(u) = (\tau + D\kappa^2,\ \xi - i\kappa).$$

This is the ideal of one point of $V(P)$, strictly larger than (P). Indeed P belongs to it:

$$P = (\tau + D\kappa^2) - D\,(\xi - i\kappa)(\xi + i\kappa).$$

A finite superposition. For $u = \sum_{j=1}^{r} c_j e^{-D\kappa_j^2 t + i\kappa_j x}$ with distinct κ_j, $u \in \mathrm{EP}$. Its annihilator is the ideal of the r points $(-D\kappa_j^2,\ i\kappa_j)$. It contains many operators that are not multiples of P.

The heat kernel. Let $u = t^{-1/2} e^{-x^2/(4Dt)}$ for $t > 0$. Its x-derivatives are polynomials of growing degree times the same Gaussian. They span an infinite-dimensional space, so $u \notin \mathrm{EP}$ (Lemma 3.3). Hence $\mathrm{Ann}(u) = (P)$: every constant-coefficient equation satisfied by the heat kernel is a multiple of $\tau - D\xi^2$.

Message. A "poor" solution remembers only finitely many exponentials and cannot tell the heat equation from many others. A "rich" solution remembers the whole equation.

Example 4.4 The wave equation and d'Alembert (C3-T5) [proved]

Let $P = (\tau - c\xi)(\tau + c\xi)$ with real $c \neq 0$, and $u = f(x - ct) + g(x + ct)$. The two partial quotients are

$$w_1 = (\partial_t + c\partial_x)\,u = 2c\,g'(x + ct),$$
$$w_2 = (\partial_t - c\partial_x)\,u = -2c\,f'(x - ct).$$

So the criterion says:

$$\mathrm{Ann}(u) = (P) \iff f \notin \mathrm{EP} \ \text{ and } \ g \notin \mathrm{EP}.$$

A single travelling wave ($g = 0$) never identifies the wave equation: it also satisfies the first-order transport equation. The same theorem covers Klein–Gordon $\tau^2 - \xi^2 + \mu^2$ and the telegraph operator $\tau^2 + \alpha\tau - c^2\xi^2$: both are irreducible conics, so they are identified iff $u \notin \mathrm{EP}$. For the beam $\tau^2 + \xi^4 = (\tau - i\xi^2)(\tau + i\xi^2)$ both Schrödinger parts must be outside EP.

Dim. C3-T5, Catalogue p. 64

Why it matters. Theorem 4.2 turns "does u identify P?" into finitely many membership tests "$\in$ EP". Each test is a finite-dimensional computation (Lemma 3.3). Whether the tests can be carried out from the data one actually has is the subject of Chapter 5.

4.3 The intrinsic operator and the spectral form

Theorem 4.2 says when (P) is the whole annihilator. When it is not, what does u determine? The answer uses the two pieces of $\mathrm{Ann}(u)$ given by Lemma 3.2.

Point spectrum

Recall $\mathrm{Ann}(u) = G_u J_u$ with $G_u = \gcd \mathrm{Ann}(u)$. The *point spectrum* of u is the finite set

$$\sigma_p(u) = V(J_u) \subset \mathbb{C}^2.$$

Its points are the exponents (τ_0, ξ_0) of the exponentials $e^{\tau_0 t + \xi_0 x}$ that are "present" in u. The word spectrum is meant in the algebraic sense, not as the spectrum of an operator on a Hilbert space. We write $G_u \sim P$ when the two agree up to a nonzero constant.

Theorem 4.5 Spectral characterisation (P4-T2) [proved]

Let P be an annihilator of u. Then

$$\mathrm{Ann}(u) = (P) \iff \sigma_p(u) = \emptyset \text{ and } G_u \sim P.$$

Moreover the exponential-polynomial part of the derivative space is the finite quotient

$$\mathcal{R}u \cap \mathrm{EP} \cong \mathcal{R}/J_u.$$

Dim. P4-T2, Catalogue p. 10

Example 4.6 Reading the spectrum

One exponential. For $u = e^{\tau_0 t + \xi_0 x}$: $\mathrm{Ann}(u) = (\tau - \tau_0,\ \xi - \xi_0)$, so $G_u = 1$ and $\sigma_p(u) = \{(\tau_0, \xi_0)\}$. A single exponential has no infinite part and a one-point spectrum.

Two exponentials. For $u = e^{\tau_1 t + \xi_1 x} + e^{\tau_2 t + \xi_2 x}$ with distinct exponents: $G_u = 1$ and $\sigma_p(u)$ has two points.

A rich solution. If $\mathrm{Ann}(u) = (P)$ then $\sigma_p(u) = \emptyset$ and $G_u \sim P$.

The operator G_u is the correct "operator intrinsic to u". It depends only on u, while P depends on which annihilator was chosen. The next theorem computes it from any annihilator.

Theorem 4.7 Intrinsic operator (C3-T6) [proved]

Let $P = c\prod_i P_i^{e_i}$ be any annihilator of u. For each i let

$$k_i = \max\left\{k \le e_i : \left(\frac{P}{P_i^k}\right)(\partial)u \in \mathrm{EP}\right\}$$

(with $k_i = 0$ if no such $k \ge 1$ exists), and $f_i = e_i - k_i$. Then

$$G_u \sim \prod_i P_i^{f_i},$$

and $\mathrm{Ann}(u)$ is principal if and only if $G_u(\partial)u = 0$. Computing G_u takes at most $\sum_i e_i$ tests "$\in$ EP".

Dim. C3-T6, Catalogue p. 9

Example 4.8 A repeated factor

Let G be the heat kernel ($D = 1$), $P = (\tau - \xi^2)^3$ and $u = t\,G$. Then:

$$\left(\frac{P}{P_1}\right)(\partial)u = (\partial_t - \partial_x^2)^2(tG) = 0 \in \mathrm{EP},$$

$$\left(\frac{P}{P_1^2}\right)(\partial)u = (\partial_t - \partial_x^2)(tG) = G \notin \mathrm{EP}.$$

So $k_1 = 1$, $f_1 = 2$, and

$$G_u = (\tau - \xi^2)^2, \qquad \mathrm{Ann}(u) = (G_u).$$

The function tG identifies the *square* of the heat operator, not its cube.

Why G_u and not the variety

A tempting stronger statement replaces G_u by the variety $V(\mathrm{Ann}\,u)$. It is correct only for reduced P, because zero sets do not see multiplicities.

Example. Let $P = \tau^2$ and $u = e^{x^2}$. Then $u_{tt} = 0$. No nonzero polynomial in ∂_x alone kills u, so

$$\mathrm{Ann}(u) = (\tau).$$

Here $\sigma_p(u) = \emptyset$, but $G_u = \tau \not\sim P$. So $\mathrm{Ann}(u) \neq (P)$, while $V(\mathrm{Ann}\,u) = \{\tau = 0\} = V(P)$. The variety alone would wrongly say that u identifies τ^2.

4.4 Refinements

4.4.1 Evolutions: identification inside a finite class

For an evolution $u_t = a(\partial_x)u$ the symbol $P = \tau - a(\xi)$ has degree one in τ, so it is irreducible. The criterion says: $\mathrm{Ann}(u) = (P)$ iff $u \notin \mathrm{EP}$. When u *is* an exponential-polynomial, we can still ask whether a is recovered among polynomials of degree at most n.

Theorem 4.9 Evolutions (C3-T4) [proved]

Let $P = \tau - a(\xi)$ and $P(\partial)u = 0$ on a connected Ω.

(a) $\mathrm{Ann}(u) = (P)$ if and only if $u \notin \mathrm{EP}(\Omega)$.

(b) If $u = \sum_k q_k\, e^{\tau_k t + \xi_k x}$ with distinct exponents and nonzero q_k, put $\delta_k = \deg_x q_k$ and

$$D = \sum_k (\delta_k + 1).$$

If $\deg a \le n$, then among all b of degree at most n, the evolutions $\tau - b$ that kill u are exactly

$$b \in a + \prod_k (\xi - \xi_k)^{\delta_k + 1} \cdot \mathbb{C}[\xi]_{\le n - D}.$$

So a is identified in degree $\le n$ if and only if $D \ge n + 1$.

Dim. C3-T4, Catalogue p. 64

This is Hermite interpolation: a polynomial of degree n is fixed by $n+1$ conditions, and each mode $q_k\, e^{\tau_k t + \xi_k x}$ supplies $\delta_k + 1$ of them.

Example 4.10 Advection–diffusion from two modes

Take $u_t = -cu_x + Du_{xx}$, so $a(\xi) = -c\xi + D\xi^2$ has degree $n = 2$. Two simple modes give $D = 2 < 3$: the pair (c, D) is not identified, because one free parameter remains. Three distinct modes give $D = 3$: both coefficients are fixed.

4.4.2 A naive transfer that fails

On a periodic domain an earlier result said: if the Fourier expansion of u has infinitely many frequencies on every factor curve, then $\mathrm{Ann}(u) = (P)$ (R4-T5). On a *bounded* domain this transfer is false, because the expansion is not unique.

Counterexample 4.11 The spectrum of an expansion is not intrinsic (C3-C1)

Let $P = \tau + \xi$ and $\Omega = (0,1)^2$. On $(-\pi, \pi)$ the Fourier series

$$e^s = \sum_{k \in \mathbb{Z}} c_k\, e^{iks}, \qquad c_k = \frac{\sinh(\pi)(-1)^k}{\pi(1 - ik)} \neq 0,$$

converges in L^2. So $u = e^{x-t} = \sum_k c_k\, e^{ik(x-t)}$ has infinitely many exponents on the line $\tau + \xi = 0$. Yet

$$\mathrm{Ann}(u) = (\tau + 1,\ \xi - 1) \supsetneq (\tau + \xi).$$

Dim. C3-C1, Catalogue p. 65

The criterion must therefore be stated on u itself, as in Theorem 4.2, not on an expansion of u.

4.4.3 Symmetries and local data

Proposition 4.12 Translation symmetry (C3-P3) [proof written, not audited]

Suppose u is invariant under translations along a direction, $\ell(\partial)u = 0$ with $\ell = \alpha\tau + \beta\xi \neq 0$, and $P(\partial)u = 0$ with P nonconstant. If $P \not\sim \ell$, then $\mathrm{Ann}(u) \neq (P)$. If moreover $\ell \nmid P$, then $u \in \mathrm{EP}$.

Dim. C3-P3, Catalogue p. 66

So a symmetry with constant characteristic destroys identification, and it is a special case of the criterion. Symmetries with variable coefficients are invisible in $\mathbb{C}[\partial]$: the heat kernel is scale invariant, $(2t\partial_t + x\partial_x + 1)G = 0$, yet $\mathrm{Ann}(G) = (\tau - \xi^2)$. Identifiability depends on the ring of coefficients.

Proposition 4.13 Local data (C3-P5) [proof written, not audited]

Identification can be lost on a smaller region. Let $P = \tau + \xi$, $F(s) = e^s + \varphi(s)$ with φ a smooth bump supported in $(2,3)$, and $u = F(x-t)$. On $(0,1)\times(-5,5)$, $u \notin \mathrm{EP}$ and $\mathrm{Ann}(u) = (\tau + \xi)$. On the smaller square $(0,1)^2$, $u = e^{x-t}$ and the annihilator is the ideal of a point.

For elliptic P, every solution is real-analytic, and then $\mathrm{Ann}(u|_{\Omega'}) = \mathrm{Ann}(u)$ for every connected $\Omega' \subset \Omega$.

Dim. C3-P5, Catalogue p. 67

Takeaway. A solution identifies its equation exactly when, after removing any one irreducible factor, what is left is not a finite sum of exponentials. In general, u identifies only its intrinsic operator G_u. The difference between P and G_u is a finite set of exponentials, the point spectrum $\sigma_p(u)$.

CHAPTER 5

Finite data and decidability

In this chapter	
Starting from	the criterion of Chapter 4: $\mathrm{Ann}(u) = (P)$ iff no w_i is an exponential-polynomial. It is a statement about the whole function u.
Why	in practice we hold finitely many numbers: a jet at a point, samples, or a formula. We ask which data suffice, which never suffice, and when the test "$\in$ EP" becomes an algorithm.
Role	tells Step 3 of the algorithm what it can and cannot promise. The answer depends on the data model: positive for exponential-polynomials, negative for a black box, positive again with one extra integer.

5.1 Three data models

The reconstruction is exact, so the input must be of an exact kind. Three models are analysed throughout (C3-D1).

M1 A closed formula for u, for example a finite sum of exponentials and Gaussians.

M2 The jet of u at a point p_0: all partial derivatives up to some order, given as exact numbers. In the "black-box" version M2$^+$ we may also ask a *zero-test oracle* whether $Q(\partial)u = 0$ for a given Q.

M3 u real-analytic on a connected Ω, described by finitely many linear equations with polynomial coefficients (a *holonomic* description) with algebraic numbers as coefficients.

Words used in this chapter

Jet of order k	the list of all partial derivatives of u of order $\le k$ at p_0: its Taylor coefficients.
Zero-test oracle	a procedure that says whether a given expression built from u is identically zero.
Holonomic	described by linear differential equations with polynomial coefficients whose solution space is finite-dimensional.
Weyl algebra	A_1 or A_2: the ring of differential operators with polynomial coefficients.

5.2 Exponential-polynomial data: a finite threshold

When u itself is an exponential-polynomial, it never identifies its equation in the whole ring: its annihilator has finitely many zeros, while $V(P)$ is a curve (C3-C2). What we can ask is whether u identifies P *among operators of degree at most* n.

Theorem 5.1 Bézout threshold (C3-T2) [proved]

Let $u = \sum_k c_k\, e^{\langle z_k,(t,x)\rangle}$ be a sum of simple modes with distinct exponents $z_k \in V(P)$, P reduced, and $n \geq \deg P$. If every factor curve carries enough exponents,

$$\big|\{z_k\} \cap V(P_i)\big| > n \cdot \deg P_i \qquad \text{for every } i,$$

then the annihilators of degree at most n are exactly the multiples of P:

$$\mathrm{Ann}(u) \cap \mathcal{R}_{\leq n} = P \cdot \mathcal{R}_{\leq n-\deg P}.$$

The threshold cannot be lowered uniformly: for $P = \tau - \xi^2$ and every n there are $2n$ exponents on the parabola that do not suffice.

Dim. C3-T2, Catalogue p. 12

Intuition

An operator Q of degree n that kills u must vanish at every exponent z_k. If P_i does not divide Q, the two curves meet in at most $n \deg P_i$ points (Bézout). More points than that force P_i to divide Q.

Example 5.2 The parabola needs $2n+1$ points

Let $P = \tau - \xi^2$, so $\deg P = 2$. Take the $2n$ exponents

$$Z_n = \{(j^2, \pm j) : \ j = 1, \dots, n\} \subset V(P).$$

The operator $Q_n = \prod_{j=1}^{n} (\tau - j^2)$ has degree n, vanishes on Z_n, and is not a multiple of $\tau - \xi^2$ because it does not depend on ξ. So $2n$ points are not enough. With $2n+1$ points the theorem applies.

The next theorem says how many derivatives at a single point are needed.

Theorem 5.3 Jet threshold $2d-1$ (C3-T3) [proved]

Let $0 \neq u \in \mathrm{EP}(\Omega)$ on a connected Ω, and let $d = \dim \mathcal{R}u$. Then:

(i) the operators of degree at most n that kill u are determined by the jet of u of order $n + d - 1$ at any point;

(ii) $\mathrm{Ann}(u)$ is generated in degree at most d, so it is determined by the jet of order $2d - 1$;

(iii) the order $2d - 1$ is optimal in the worst case, already in one variable: for $u = \sum_{j=1}^{d} e^{jx}$ the jet of order $2d - 2$ does not determine $\mathrm{Ann}(u)$.

An upper bound $d' \geq d$ may replace d everywhere.

Dim. C3-T3, Catalogue p. 12

Example 5.4 One exponential, $d = 1$

For $u = e^{2x}$, $d = 1$ and the threshold is $2d - 1 = 1$. The values $u(0) = 1$ and $u'(0) = 2$ give $u' = 2u$, so $\mathrm{Ann}(u)$ contains $\xi - 2$, and τ comes from the absence of t. The value $u(0)$ alone (order 0) cannot distinguish e^{2x} from e^{3x}.

5.3 What no finite data can show

For a solution outside EP, the situation reverses.

Theorem 5.5 No finite certificate (C3-P1) [proved] / [from cited source]

Let Ω be connected, P nonconstant and $u \in \ker P$. For every order K there is an exponential-polynomial $v \in \ker P$ with the same jet of order K at p_0 as u, and $\mathrm{Ann}(v) \neq (P)$.

On convex Ω the same holds for any finite family of continuous linear functionals (Malgrange's density theorem) [from cited source]. On a non-convex region a single functional can certify.

Dim. C3-P1, Catalogue p. 14

Intuition

Every finite jet is matched by a finite sum of exponentials taken from the curve $V(P)$. Finitely many numbers cannot distinguish u from that sum, and the sum never identifies anything. So an algorithm that sees finitely many numbers can only *refute* identification, never confirm it.

The black-box version is sharper.

Theorem 5.6 Black-box impossibility (P4-T5) [proved]

In the model $\mathrm{M2}^+$ (exact jets plus a zero-test oracle), for every nonconstant P there is a solution u with $\mathrm{Ann}(u) = (P)$ such that the following holds. For every jet order k_0 and every finite set of zero tests, some exponential-polynomial $v \in \ker P$ gives the same jet and the same answers, while $\mathrm{Ann}(v) \neq (P)$. Hence no algorithm decides "$\mathrm{Ann}(u) = (P)$" correctly on all inputs.

Dim. P4-T5, Catalogue p. 16

5.4 One extra integer makes it decidable

Theorem 5.7 Decidability with a bound (P4-T4) [proved]

In the model M2$^+$:

(i) for every finite-dimensional space of operators $\mathcal{L}$, the annihilators of u in $\mathcal{L}$ are computable;

(ii) "$\mathrm{Ann}(u) \neq (P)$" is semi-decidable: an algorithm stops with "yes" exactly when it is true;

(iii) if an integer $B \geq \dim(\mathcal{R}u \cap \mathrm{EP})$ is given, "$\mathrm{Ann}(u) = (P)$" is decidable. With

$$N = B + \deg P - 1,$$

one has $\mathrm{Ann}(u) = (P)$ iff $\mathrm{Ann}(u) \cap \mathcal{R}_{\leq N} = (P) \cap \mathcal{R}_{\leq N}$.

The bound also makes G_u computable.

Dim. P4-T4, Catalogue p. 15

Intuition

If identification fails, some w_i is an exponential-polynomial of dimension at most B. Its annihilator has a generator of degree at most B that is not a multiple of P_i. Multiplying by P/P_i gives a witness of degree at most $B + \deg P - 1$. So it is enough to look for witnesses up to that degree, and that is a finite linear-algebra problem.

Is the bound B the least data that works? The minimal sufficient datum is "a degree D such that, if identification fails, a witness of degree $\leq D$ exists" (OP-ALG-T3) [proof written, not audited]. The bound B is strictly stronger on some regions when P has at least two irreducible factors counted with multiplicity. Whether it is strictly stronger on every region is open [open] (OP-ALG-Q2).

5.5 A decidable class: holonomic solutions

Theorem 5.8 Holonomic model (OP-ALG-T1, P4-T6) [proved mod. (AC)]

In the model M3 (real-analytic u with a holonomic description over algebraic numbers), assume (AC): jets at an algebraic point are computable in a computable field. Then

- "$Q(\partial)u \in \mathrm{EP}$?" is decidable for every Q;
- "$\mathrm{Ann}(u) = (P)$?" and "$V(\mathrm{Ann}\, u) = V(P)$?" are decidable, and G_u is computable;
- a bound B and a finite set of generators of $\mathrm{Ann}(u)$ are computable.

Dim. OP-ALG-T1, Catalogue p. 68

Example 5.9 The heat kernel plus an exponential

Let $u = G + e^{t+x}$, with G the heat kernel. The slice $t = 1$ is $x \mapsto G(1,x) + e^{1+x}$, a function with a holonomic description. The algorithm finds that the slice is not an exponential-polynomial, and concludes $\mathrm{Ann}(u) = (\tau - \xi^2)$. For the slice of G alone the bound is $B = 0$: the heat kernel identifies the heat equation without further computation.

5.6 Variable coefficients: the first limits

With coefficients that depend on x, the ring of operators becomes the Weyl algebra. Two of its features break the constant-coefficient picture.

Proposition 5.10 Restriction does not identify (P4-C1) [proof written, not audited]

For every nonzero smooth u of one variable, the action of the Weyl algebra A_1 on the module $A_1 \cdot u$ is faithful. Yet for $u = e^x$ and the class $\mathcal{L} = \mathrm{span}\{1, \partial\}$, the operator $\partial - 1$ kills u. So "the restriction is injective" does not detect identification. The correct object is the evaluation map $L \mapsto Lu$, whose kernel is $\mathrm{Ann}(u) \cap \mathcal{L}$.

Dim. P4-C1, Catalogue p. 68

Theorem 5.11 Saturated criterion (OP-ALG-T4) [proof written, not audited]

Let P be monic in ∂_t of degree m, with coefficients in a ring $\mathfrak{A}$ of operators in x, and $Pu = 0$. Then u identifies P if and only if the t-jet $u, \partial_t u, \ldots, \partial_t^{m-1} u$ is free over $\mathfrak{A}$: no nontrivial relation $\sum_{j<m} b_j \partial_t^j u = 0$ with $b_j \in \mathfrak{A}$.

Dim. OP-ALG-T4, Catalogue p. 70

In the commutative case this is Proposition 3.4.3. With variable coefficients the reduction of the test to the finitely many quotients P/P_i fails: an explicit counterexample with $\theta = x\partial_x$ exists (OP-ALG-C1) [proof written, not audited]. In the full Weyl algebra A_2, a holonomic solution never identifies its equation (OP-ALG-P5) [proof written, not audited].

5.7 Summary

Data	What can be done	Meaning for the algorithm	Status
$u \in \mathrm{EP}$	jet of order $2d-1$ determines $\mathrm{Ann}(u)$; Bézout threshold for modes	a finite computation with a guarantee	[proved]
$u \notin \mathrm{EP}$, finite data	no finite certificate	output "not refuted", never "identified"	[proved] / [from cited source]
black box	"$\in$ EP?" undecidable	extra hypothesis needed	[proved]
bound B given	decidable, test degree $B + \deg P - 1$	stops with a certain verdict	[proved]
holonomic, algebraic	decidable	exact symbolic route	[proved mod. (AC)]
variable coefficients	restriction never identifies; jet freeness criterion	restrict to a finite class	[proof written, not audited]

Table 5.1: When the identification test can be carried out, and what each case means for Step 3 of the algorithm.

Takeaway. Identification is an infinite statement about u. It cannot be verified from finitely many numbers, and no black box can decide it. One extra integer, a bound on how much of u is exponential-polynomial, turns it into a finite computation.

CHAPTER 6

Decomposition into components

In this chapter

Starting from	the operator is known and factored: $P = c\prod_{j=1}^{s} Q_j$ with $Q_j = P_j^{m_j}$ pairwise coprime. For the main theorem Ω is open, connected and simply connected (more precisely $H^1_{dR}(\Omega) = 0$).
Why	once P is known, we want to split u into the parts generated by each piece of P, so that the pieces can be studied separately.
Role	second link of the chain. The split exists for *factors* of the symbol, with a finite and exactly known ambiguity. It fails for additive *terms*, and that failure is what Chapter 8 measures.

6.1 Which pieces? Terms versus factors

An operator can be broken up in two different ways. Take the heat operator

$$L = \partial_t - D\partial_x^2.$$

Its *terms* are $L_1 = \partial_t$ and $L_2 = -D\partial_x^2$: they are added. Its symbol $\tau - D\xi^2$ is irreducible, so it has only one *factor*. For the wave operator the situation is the opposite: $\tau^2 - \xi^2$ has two terms and also two factors, $\tau - \xi$ and $\tau + \xi$. The first question is which kind of piece gives a decomposition of the solution.

6.1.1 Terms do not give components

Proposition 6.1 Splitting by terms fails (R2-T3) [proved]

Let $L = L_1 + L_2$ with constant coefficients. If $u = a + b$ with $L_1 a = 0$ and $L_2 b = 0$ solves $Lu = 0$, then u is also killed by $L_1 L_2$:

$$\ker L \cap (\ker L_1 + \ker L_2) \;\subseteq\; \ker(L,\ L_1 L_2).$$

The common zero set of L and $L_1 L_2$ is $V(L_1) \cap V(L_2)$. When this set is finite, only a finite-dimensional space of exponential-polynomial solutions can be split by terms.

Dim. R2-T3, Catalogue p. 19

Intuition

The two operators commute, so

$$L_1 L_2\, u = L_2(L_1 a) + L_1(L_2 b) = 0.$$

A solution split by terms satisfies one extra equation. A generic solution does not.

Example 6.2 The heat equation cannot be split by terms

Take $L_1 = \partial_t$ and $L_2 = -D\partial_x^2$. The kernels are

$$\ker L_1 = \{f(x)\}, \qquad \ker L_2 = \{a(t) + b(t)\, x\}.$$

Every element $f(x) + a(t) + b(t)x$ of the sum is killed by $\partial_t \partial_x^2$. A heat solution is not. For $u = e^{-D\kappa^2 t} \cos \kappa x$:

$$\partial_x^2 u = -\kappa^2 u, \qquad \partial_t \partial_x^2 u = D\kappa^4 u \neq 0.$$

So a generic heat solution is *not* a sum of a solution of $u_t = 0$ and a solution of $u_{xx} = 0$.

The exceptions are few. The function $u = \frac{1}{2}x^2 + Dt$ solves the heat equation and lies in $\ker L_1 + \ker L_2$, but it is a polynomial: the exceptional space is finite-dimensional.

6.1.2 Factors do give components

Example 6.3 The wave equation and d'Alembert

Let $c > 0$ and

$$P = \tau^2 - c^2\xi^2 = (\tau - c\xi)(\tau + c\xi) = Q_1 Q_2.$$

A direct check gives

$$(\partial_t - c\partial_x)\, f(x + ct) = 0, \qquad (\partial_t + c\partial_x)\, g(x - ct) = 0.$$

So $\ker Q_1 = \{f(x+ct)\}$ (profiles moving left) and $\ker Q_2 = \{g(x-ct)\}$ (profiles moving right). D'Alembert's formula says that every solution is a sum of the two:

$$\ker P = \ker Q_1 + \ker Q_2.$$

The pieces are unique up to a constant: $f \mapsto f + k$, $g \mapsto g - k$. This ambiguity has dimension 1, and it equals $\dim \mathcal{R}/(Q_1, Q_2)$:

$$Q_1 + Q_2 = 2\tau, \qquad Q_2 - Q_1 = 2c\xi \quad \Longrightarrow \quad (Q_1, Q_2) = (\tau, \xi), \qquad \mathcal{R}/(\tau, \xi) \cong \mathbb{C}.$$

6.2 Why factors work: the Chinese remainder theorem

One variable. If two polynomials Q_1, Q_2 in one variable are coprime, there are polynomials a, b with

$$a\, Q_1 + b\, Q_2 = 1.$$

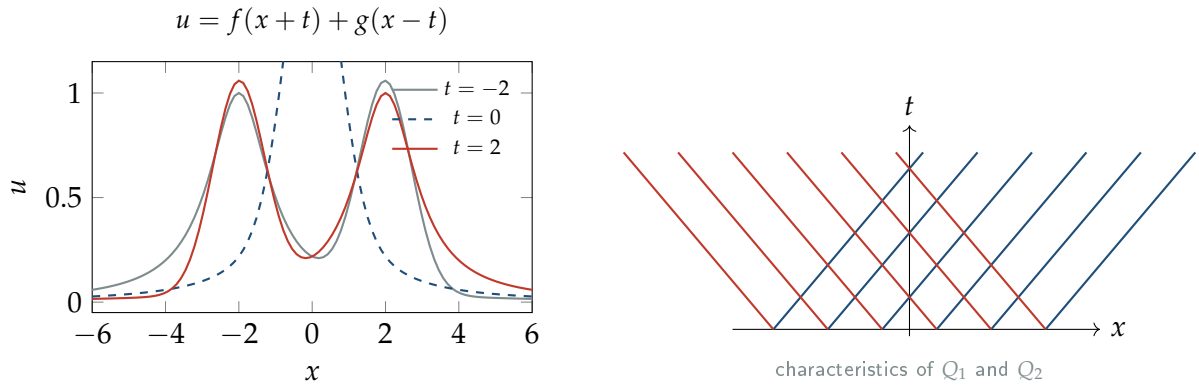


Figure 6.1: Left: the d'Alembert solution with $f(s) = 1/(1+s^2)$ moving left and $g(s) = e^{-s^2}$ moving right ($c = 1$). Right: the two families of characteristic lines, one per factor. Each factor of the symbol carries one rigidly translating component.

Then every u with $Q_1 Q_2 u = 0$ splits explicitly as

$$u = \underbrace{b\, Q_2\, u}_{\in \ker Q_1} + \underbrace{a\, Q_1\, u}_{\in \ker Q_2},$$

and the sum of the two kernels is direct.

Two variables. "Coprime" now means only that the common zeros are finitely many (Bézout), not that $(Q_1, Q_2) = \mathcal{R}$. The finite quotient $\mathcal{R}/(Q_1, Q_2)$ measures the failure of the one-variable picture.

Lemma 6.4 Finite quotient (C12-L3) [proved]

Let $f, g, h \in \mathcal{R} \setminus \{0\}$.

(a) If f, g are coprime, the ideal (f, g) contains a nonzero polynomial in ξ alone and a nonzero polynomial in τ alone. Hence $\dim \mathcal{R}/(f, g) < \infty$.

(b) If f is coprime to g and to h, then

$$\dim \mathcal{R}/(f, gh) = \dim \mathcal{R}/(f, g) + \dim \mathcal{R}/(f, h).$$

(c) $(f, g) = \mathcal{R}$ if and only if $V(f) \cap V(g) = \emptyset$.

Dim. C12-L3, Catalogue p. 21

The geometric meaning of $\dim \mathcal{R}/(Q_i, Q_j)$ is the number of crossing points of the curves $V(Q_i)$ and $V(Q_j)$, counted with intersection multiplicity.

6.3 The decomposition theorem

Theorem 6.5 Factor decomposition (C12-T1, C12-T1′) [proved]

Let $\Omega \subset \mathbb{R}^2$ be open, connected, with $H^1_{dR}(\Omega) = 0$ (for example simply connected: a rectangle, a convex set, a slit disc). Let $\mathcal{A}$ be $\mathcal{D}'(\Omega)$ or $C^\infty(\Omega)$, and $P = c\prod_{j=1}^{s} Q_j$ with pairwise coprime Q_j. Then:

(i) every solution splits:

$$\ker_{\mathcal{A}} P = \ker_{\mathcal{A}} Q_1 + \cdots + \ker_{\mathcal{A}} Q_s;$$

(ii) the ambiguity (the s-tuples of components with sum 0) has dimension exactly

$$\delta(P) = \sum_{i<j} \dim \mathcal{R}/(Q_i, Q_j),$$

and consists of exponential-polynomials;

(iii) the splitting is unique if and only if $V(Q_i) \cap V(Q_j) = \emptyset$ for all $i \neq j$.

Dim. C12-T1p, Catalogue p. 23

Intuition

At a crossing point z of two factor curves, the exponential $e^{\langle z,(t,x)\rangle}$ belongs to both kernels. It can be moved freely from one component to the other. The number of such free moves, counted with multiplicity, is $\delta(P)$.

Algebraically the proof uses the exact sequence

$$0 \to \mathcal{R}/(Q_1Q_2) \to \mathcal{R}/(Q_1) \oplus \mathcal{R}/(Q_2) \to \mathcal{R}/(Q_1, Q_2) \to 0$$

and shows that applying $\mathrm{Hom}(\,\cdot\,, \mathcal{A})$ keeps it exact. The only possible obstruction lives on the finite-dimensional module $\mathcal{R}/(Q_1, Q_2)$, and it vanishes when Ω has no holes (Lemma 6.3).

Example 6.6 The ambiguity on four operators

Operator	Factors	$\delta(P)$	Ambiguity
wave $\tau^2 - \xi^2$	$\tau - \xi$, $\tau + \xi$	1	constants
beam $\tau^2 + \xi^4$	$\tau - i\xi^2$, $\tau + i\xi^2$	2	span$\{1, x\}$
heat–transport	$\tau - \xi^2$, $\tau - \xi$	2	span$\{1,\ e^{t+x}\}$
parallel transports	$\tau - \xi$, $\tau - \xi - 1$	0	none (unique)

For heat–transport the curves $\tau = \xi^2$ and $\tau = \xi$ meet at $(0,0)$ and $(1,1)$, so the shared exponentials are 1 and e^{t+x}. For the two parallel transports the lines $\tau = \xi$ and $\tau = \xi + 1$ never meet, and the splitting is unique.

The key analytic tool is a version of the Poincaré lemma "closed forms are exact" with an exponential weight.

Lemma 6.7 Twisted Poincaré lemma (C12-L1′, L2′) [proved]

Let Ω be connected with $H^1_{dR}(\Omega) = 0$, and $z = (z_1, z_2) \in \mathbb{C}^2$. Write $D_t = \partial_t - z_1$ and $D_x = \partial_x - z_2$. If $\varphi, \psi \in \mathcal{A}$ satisfy the compatibility condition

$$D_x \varphi = D_t \psi,$$

then there is $w \in \mathcal{A}$ with $D_t w = \varphi$ and $D_x w = \psi$. Consequently, for every finite-dimensional $\mathcal{R}$-module M, the obstruction group $\mathrm{Ext}^1_{\mathcal{R}}(M, \mathcal{A})$ vanishes.

Dim. C12-L1p, Catalogue p. 22

Multiplying by $e^{-z_1 t - z_2 x}$ turns the twisted derivatives into ordinary ones, and the lemma becomes the classical statement that a closed 1-form on a simply connected region is exact.

6.4 Where the decomposition fails: holes

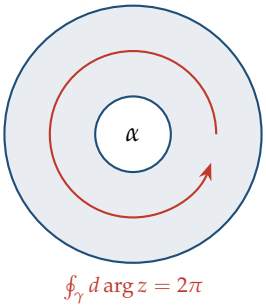
Proposition 6.8 Holes break it (C12-P10) [proved]

Let Ω contain a closed curve γ that winds around a point $\alpha \notin \Omega$. For the Laplacian

$$P = \tau^2 + \xi^2 = (\tau - i\xi)(\tau + i\xi),$$

the harmonic function $u = \log|(t, x) - \alpha|^2$ is *not* a sum of a solution of $\partial_t - i\partial_x$ and a solution of $\partial_t + i\partial_x$.

Dim. C12-P10, Catalogue p. 25



$u = \log|z - \alpha|^2$ is harmonic on the annulus, but $u \neq h(z) + \overline{g(z)}$ with h, g single-valued holomorphic: the logarithm has monodromy.

Figure 6.2: The Laplacian factors into two Cauchy–Riemann operators, but on an annulus the factor decomposition fails. The obstruction is topological: $H^1_{dR}(\Omega) \neq 0$.

For a fixed P the exact condition is cohomological.

Theorem 6.9 Cohomological criterion (OPF-T4) [proof written, not audited]

Let Ω be connected, with first Betti number $b_1(\Omega)$. Let $\mathcal{N}$ be the space of s-tuples of global exponential-polynomial solutions with sum zero (dimension $\delta(P)$). The cokernel of the sum map

$$\Sigma_\Omega : \bigoplus_j \ker_{\mathcal{A}(\Omega)} Q_j \to \ker_{\mathcal{A}(\Omega)} P$$

is the kernel of an explicit map $H^1(\Omega;\mathbb{C}) \otimes \mathcal{N} \to \bigoplus_j H^1(\Omega;\mathcal{K}_j)$. So the decomposition holds on Ω iff that map is injective, and

$$\dim \operatorname{coker} \Sigma_\Omega \le \delta(P)\, b_1(\Omega).$$

Dim. OPF-T4, Catalogue p. 78

Example 6.10 Unique splittings survive holes

If $\delta(P) = 0$ (disjoint factor curves), the bound gives $\operatorname{coker} \Sigma_\Omega = 0$ on *every* connected region, with or without holes. The two parallel transports $\tau - \xi$ and $\tau - \xi - 1$ split on an annulus as well.

6.5 Explicit projectors

For evolution factors the components are given by a formula. Let

$$P = \prod_{j=1}^{N} (\tau - a_j(\xi))$$

with distinct polynomials a_j. A component is an eigenfunction of ∂_t:

$$\partial_t u_j = a_j(\partial_x)\, u_j.$$

Apply $\prod_{k\neq j}(\partial_t - a_k(\partial_x))$ to $u = \sum_l u_l$. Every u_l with $l \neq j$ is killed, because the product contains its own factor. The component u_j is multiplied by $\prod_{k\neq j}(a_j - a_k)(\partial_x)$. Dividing gives the Lagrange formula

$$u_j = \prod_{k\neq j} \frac{\partial_t - a_k(\partial_x)}{a_j(\partial_x) - a_k(\partial_x)}\, u. \tag{6.1}$$

In Fourier variables the right-hand side is a multiplier that is rational in ξ, with poles where two a_k coincide. This is Eq. (3.2) with the companion matrix replaced by ∂_t.

Example 6.11 Two transports

Let $a_1(\xi) = \xi$ and $a_2(\xi) = \xi + 1$. The formula gives

$$u_1 = \frac{\partial_t - \partial_x - 1}{-1}\, u, \qquad u_2 = \frac{\partial_t - \partial_x}{1}\, u.$$

These are differential operators, with no poles, because $a_1 - a_2 = -1$ never vanishes: the two lines are disjoint and the splitting is unique.

Proposition 6.12 Projectors and their poles (II-T3, II-T4) [proved]

The components are obtained by *polynomial* operators if and only if the factor curves are pairwise disjoint, $(Q_i, Q_j) = \mathcal{R}$. Only factors, not terms, admit projectors that act on one value of u at a time.

Dim. II-T3, Catalogue p. 61

Takeaway. A "component" of u belongs to a *factor* of the symbol, not to a term of the operator. On a region without holes the factor split always exists, and the overlap between components is finite-dimensional and measured exactly by $\delta(P)$. The split by terms fails except on a finite-dimensional space.

CHAPTER 7

Regularity of the components

In this chapter

Starting from	the factor decomposition of Theorem 6.3; Ω bounded; u in a Sobolev space $H^m(\Omega)$ (square-integrable derivatives up to order m).
Why	the decomposition is algebraic. To use the components numerically we need them to be as regular as u itself, and to depend continuously on u.
Role	completes the answer to Q2. It gives the exact condition (a conditioning number $\gamma_N > 0$), shows that it can fail, and that the geometry of Ω against the characteristics decides it. Step 5 of the algorithm checks it.

7.1 The pieces can be rougher than the whole

Counterexample 7.1 Loss of a derivative (C12-C1)

The statement "if $u \in H^m(\Omega)$ then every component $u_j \in H^m(\Omega)$" is false. Let $\Omega = (0,1)^2$ and

$$f = \tau - \xi, \qquad g = \tau - \xi - 1.$$

For every m there is a smooth $u \in H^m(\Omega)$ with $f(\partial)g(\partial)u = 0$ whose unique components $u_1 \in \ker f$, $u_2 \in \ker g$ are *not* in H^m. Exactly one derivative is lost: $u_2 = f(\partial)u \in H^{m-1}$.

Dim. C12-C1, Catalogue p. 26

Intuition

In the coordinates $s = x + t$ along the characteristics, both factors are first-order ODEs along the same lines, with solutions 1 and e^t. On a long chord these two functions are clearly different. Near a corner of the square the chords become arbitrarily short, and on a short chord 1 and e^t are almost collinear. Separating a function into these two shapes then amplifies it by a factor that blows up as the chord shrinks (Figure 7.1).

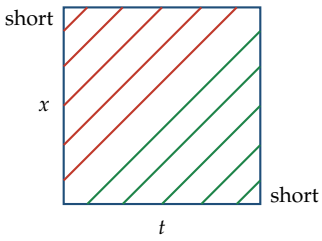


Figure 7.1: The unit square and the characteristic chords of $\partial_t + \partial_x$. Chords near the corners $(0,1)$ and $(1,0)$ are arbitrarily short. On a short chord the two shapes of the components are nearly collinear, and the splitting is ill-conditioned.

7.2 Two components: the chord criterion

For two transports in the same direction, the geometry decides everything.

Theorem 7.2 Chord criterion (C12-T3) [proved]

Let Ω be bounded, open and convex, $c \in \mathbb{R}$, $d_1 \neq d_2$, and

$$f = \partial_t + c\,\partial_x - d_1, \qquad g = \partial_t + c\,\partial_x - d_2.$$

Let $\ell(s)$ be the length of the chord of Ω along the characteristic line $x - ct = s$, and $\mathcal{H}_i = \ker_{L^2} f_i$. The following are equivalent:

(a) the chords do not shrink: $\inf_s \ell(s) > 0$;

(b) every $u \in L^2$ with $fgu = 0$ has components in L^2, with

$$\|u_1\|^2 + \|u_2\|^2 \leq C\,\|u\|^2;$$

(c) the sum $\mathcal{H}_1 + \mathcal{H}_2$ is closed in L^2;

(d) the Friedrichs cosine of the two spaces is smaller than one, $c_F(\mathcal{H}_1, \mathcal{H}_2) < 1$.

Dim. C12-T3, Catalogue p. 27

Example 7.3 Rectangles

On a rectangle $I \times J$ with $c \neq 0$, the chords in direction $(1, c)$ shrink at two corners. So the global L^2 decomposition always fails ($c_F = 1$). For $c = 0$ the chords are vertical, all of length $|I|$, and the decomposition holds.

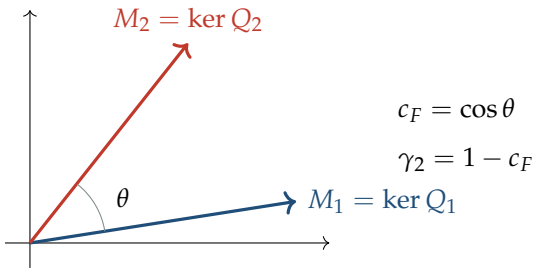


Figure 7.2: The Friedrichs angle θ between two component spaces. A larger angle means a better-conditioned recovery of the pieces from their sum. As $\theta \to 0$ the recovery becomes unstable.

7.3 Any number of components

For N components the right quantity is the conditioning of the sum map

$$S : \bigoplus_{j=1}^{N} \ker_{H^m} Q_j \to H^m, \qquad S(u_1, \dots, u_N) = \sum_{j=1}^{N} u_j.$$

Definition 7.4 Conditioning number γ_N

Let $(\ker S)^\perp$ be the tuples orthogonal to the ambiguity of Theorem 6.3. Then

$$\gamma_N = \inf_{\substack{v \in (\ker S)^\perp \\ v \neq 0}} \frac{\|Sv\|_{H^m}^2}{\|v\|^2}, \qquad \|v\|^2 = \sum_j \|u_j\|_{H^m}^2.$$

Equivalently, γ_N is the bottom of the spectrum of S^*S on $(\ker S)^\perp$: the square of the smallest nonzero singular value of S.

Intuition

γ_N is the worst ratio between the size of the sum and the size of the pieces. If $\gamma_N > 0$, the pieces cannot nearly cancel each other:

$$\|v\| \le \gamma_N^{-1/2} \, \|Sv\|,$$

and recovering them from u is stable. If $\gamma_N = 0$, there are large pieces with a tiny sum, and the recovery is unstable. It is exactly the smallest singular value of a least-squares problem.

Theorem 7.5 Regularity criterion (C12-P7) [proved]

(a) Every $u \in \ker_{H^m} P$ has components in H^m if and only if the sum $\sum_j \ker_{H^m} Q_j$ is *closed* and *dense* in $\ker_{H^m} P$.

(b) The sum is closed if and only if $\gamma_N > 0$. Then the minimal-energy split satisfies, optimally,

$$\sum_j \|u_j\|_{H^m}^2 \le \gamma_N^{-1} \, \|u\|_{H^m}^2.$$

(c) For $N = 2$, $\gamma_2 = 1 - c_F$, the Friedrichs cosine of the two spaces after removing their intersection.

(d) For $N \geq 3$ the pairwise cosines do not decide. There are three closed subspaces of a Hilbert space with trivial pairwise intersections and closed pairwise sums ($c_F \leq \frac{1}{2}$), whose triple sum is not closed.

Dim. C12-P7, Catalogue p. 29

The condition has two parts. Closedness is the conditioning above. Density is a separate question, and on convex regions it is always true.

Theorem 7.6 Density on convex regions (OPF5B-T1) [proved mod. Hörmander, Stein]

Let $\Omega \subset \mathbb{R}^2$ be bounded and convex. For every operator P, every coprime factorisation and every $m \geq 0$, the sum $\sum_j \ker_{H^m} Q_j$ is dense in $\ker_{H^m} P$.

Dim. OPF5B-T1, Catalogue p. 30

So on a convex region only $\gamma_N > 0$ remains to be checked.

Example 7.7 Cases where the H^m decomposition holds (C12-P4) [proved]

$P = \tau\xi$ on a rectangle. The components are a function of t and a function of x. After removing the constants their H^m inner product vanishes, so $c_F = 0$ and $\gamma_2 = 1$.

The wave equation on a square. In characteristic coordinates $s = x + t$, $r = x - t$ the square becomes a rhombus. It can be covered by five pieces on each of which one component is controlled by u.

Heat and transport on a rectangle (OPR2-T1) [proof written, not audited]. For $P = (\tau - \xi^2)(\tau + c\xi)$ the sum is dense in every H^m, so the decomposition holds exactly when $c_F < 1$.

The same operator can behave well on one region and badly on another: regularity depends on the *geometry of* Ω *against the characteristics*, not on P alone.

Dim. C12-P4, Catalogue p. 77

Proposition 7.8 Real transports, $m = 0$ (OPR2-T2) [proof written, not audited]

Let Ω be convex and bounded, and $Q_j = v_j \cdot \nabla - d_j$ real transports, grouped into classes of parallel directions. Assume that of any two classes at least one has chords bounded below. Then $\sum_j \ker_{L^2} Q_j$ is closed if and only if every class with at least two factors has chords bounded below. In particular, for such transports, closed pairwise sums imply a closed triple sum.

Dim. OPR2-T2, Catalogue p. 81

Three or more components: what is still open

Whether the abstract counterexample of Theorem 7.3(d) can occur for *differential* factors is open in general [open] (C12-Q2). Proposition 7.3 rules it out for real transports in L^2.

Takeaway. Algebraically the split always exists (Theorem 6.3). Analytically the components are as regular as u exactly when the component spaces are separated by a positive angle, $\gamma_N > 0$; density is automatic on convex regions. Everything delicate happens at the boundary of Ω (shrinking chords) or in its topology (holes).

CHAPTER 8

Terms inside a factor: the interaction term

In this chapter	
Starting from	an evolution factor $u_t = Au$ with $A = A_1 + \cdots + A_N$, the *terms* (advection, diffusion, dispersion, …), each a polynomial $a_n(\partial_x)$; flows $S_n(t) = e^{tA_n}$; a bounded rectangle Ω and a base time t_0. The terms commute, except in Section 8.7.
Why	Proposition 6.1.1 shows that splitting a solution by terms is never exact. Here we compute exactly what is left over, the interaction term E_{int}, how much it costs, and whether it can be computed stably.
Role	third link of the chain, answering Q3. The answer is positive in a function class of exponential type (an exact product formula, cost $O(N)$ per value) and negative in Sobolev spaces (no continuity, no stable regularisation).

8.1 From the sum of operators to the interaction term

The solution of $u_t = Au$ is given by the flow (semigroup) of A:

$$u(t) = e^{tA}u_0 = S(t)\,u_0.$$

If term n acted alone, it would move the datum to $S_n(t)u_0 = e^{tA_n}u_0$, a change of $S_n(t)u_0 - u_0$.

The naive guess is that the terms act independently, so that the total change is the sum of the individual changes. It is not: the flow of a sum is not the sum of the flows. For advection ($A_1 = -c\partial_x$, a translation) and diffusion ($A_2 = D\partial_x^2$, a smoothing), the solution is translated *and* smoothed. This joint action is not "only translated" plus "only smoothed".

Definition 8.1 Interaction term

With $S_n(t) = e^{(t-t_0)A_n}$ the flow of term n alone,

$$E_{\text{int}}^{\text{add}}(t) = u(t) - u_0 - \sum_{n=1}^{N} \big(S_n(t)u_0 - u_0\big). \tag{8.1}$$

The multiplicative variant compares the true flow with the product of the single flows:

$$E_{\text{int}}^{\text{mult}}(t) = e^{(t-t_0)A} - \prod_{n=1}^{N} e^{(t-t_0)A_n}.$$

We write E_{int} for $E_{\text{int}}^{\text{add}}$ unless the distinction matters.

8.2 The function class and the exact formula

Before computing, one must decide on which functions the flows of the single terms are defined. This is not a formality. For the heat equation there are nonzero solutions with zero initial datum (Tychonoff), so "the diffusion flow of u_0" is not unique without a class.

Definition 8.2 The class $\mathfrak{B}$ of exponential type (P1-D1)

For $\rho > 0$, $B_\rho(J)$ is the space of smooth f on the interval J with

$$\|f\|_{B_\rho}^2 = \sum_{j\geq 0} \rho^{-2j}\, \|f^{(j)}\|_{L^2(J)}^2 < \infty.$$

$\mathfrak{B}_\rho(\Omega)$ is the analogous space in (t,x), and $\mathfrak{B} = \bigcup_\rho \mathfrak{B}_\rho$. Every exponential-polynomial belongs to $\mathfrak{B}$, and on B_ρ the derivative is bounded: $\|\partial_x\| \leq \rho$.

On $\mathfrak{B}$ every polynomial in ∂_x is a bounded operator, its exponential is an absolutely convergent series, and two such exponentials commute (P1-L1, P1-L2) [proved]. So the algebra of exponentials of numbers applies to operators.

Theorem 8.3 Closed form of the interaction (P1-T2) [proved]

Let $u \in \mathfrak{B}_\rho(\Omega)$ solve $(\partial_t - a(\partial_x))u = 0$ with $a = \sum_n a_n$, and put

$$\Delta_n = e^{(t-t_0)a_n(\partial_x)} - \mathrm{Id}.$$

Then the single flows are unique in $\mathfrak{B}$, $E_{\mathrm{int}}^{\mathrm{mult}} \equiv 0$, and

$$E_{\mathrm{int}}^{\mathrm{add}}(t) = \Big[\prod_{n=1}^{N}(\mathrm{Id}+\Delta_n) - \mathrm{Id} - \sum_{n=1}^{N}\Delta_n\Big]u_0 = \sum_{|S|\geq 2}\prod_{n\in S}\Delta_n\, u_0. \tag{8.2}$$

For a datum $u_0 = \sum_k c_k e^{\zeta_k x}$, each mode is multiplied by the number

$$e_{\geq 2}(z) = \prod_n (1+z_n) - 1 - \sum_n z_n, \qquad z_n = e^{(t-t_0)a_n(\zeta_k)} - 1.$$

Dim. P1-T2, Catalogue p. 34

Intuition

Expand the product $\prod_n(\mathrm{Id}+\Delta_n)$. It gives one monomial per subset S of terms. The empty subset gives Id, the singletons give the individual changes Δ_n, and everything else is interaction. So the interaction is the sum of products of *at least two* increments: it appears only when at least two terms act together.

For two terms the formula is simply

$$E_{\mathrm{int}} = \Delta_1\Delta_2\, u_0 = (S_1 - \mathrm{Id})(S_2 - \mathrm{Id})\, u_0.$$

In each Fourier mode, the interaction is the product of the two single-term responses.

Example 8.4 Advection–diffusion with a Gaussian datum

Take $u_t = -cu_x + Du_{xx}$ and $u_0 = g(x;\sigma^2)$, where

$$g(x;v) = \frac{e^{-x^2/2v}}{\sqrt{2\pi v}}.$$

The two flows are a translation and a Gaussian smoothing:

$$S_1 u_0 = g(x-ct;\sigma^2), \qquad S_2 u_0 = g(x;s^2), \qquad u = g(x-ct;s^2),$$

with $s^2 = \sigma^2 + 2Dt$. The interaction term is

$$E_{\text{int}} = g(x-ct;s^2) - g(x-ct;\sigma^2) - g(x;s^2) + g(x;\sigma^2).$$

Figure 8.1(a) shows that it is comparable in size to u itself: it is not a small correction. In Fourier variables

$$\frac{|\widehat{E_{\text{int}}}(\kappa)|}{|\hat{u}_0(\kappa)|} = \left|e^{-ic\kappa t} - 1\right| \cdot \left|e^{-D\kappa^2 t} - 1\right| \le 2.$$

It vanishes at $\kappa = 0$ and whenever $c\kappa t \in 2\pi\mathbb{Z}$, where the translation returns to the identity.

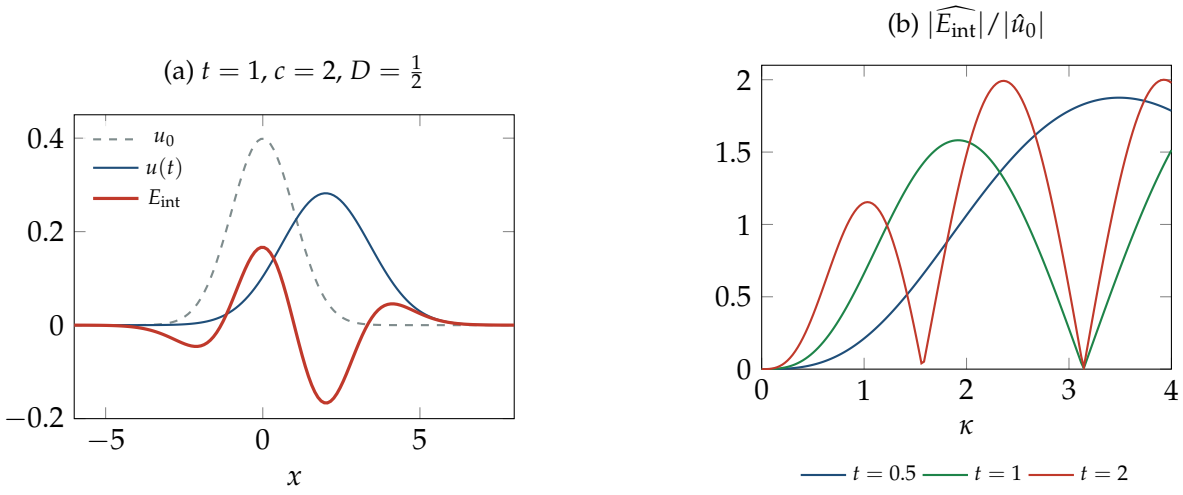


Figure 8.1: Advection–diffusion with Gaussian datum ($\sigma^2 = 1$). (a) The datum, the solution and the interaction term. (b) The Fourier ratio of the interaction to the datum, for three times.

Example 8.5 A stationary state is pure interaction

For advection–diffusion, a stationary solution satisfies $-cu_x + Du_{xx} = 0$, for example $u = e^{cx/D}$. The full flow leaves it unchanged, so $u(t) - u_0 = 0$. The single flows do change it, and the interaction term exactly cancels their changes. In this mode

$$E_{\text{int}} = -4\sinh^2\left(\frac{\theta}{2}\right) u, \qquad \theta = \frac{c^2(t-t_0)}{D}.$$

Equilibrium is the case where all the dynamics is interaction.

8.3 Cost: $O(N)$ versus 2^N

Expanding (8.2) produces one monomial per subset with at least two elements: $2^N - N - 1$ cross terms (Figure 8.2). Writing the *complete table* of cross terms is therefore exponential. Computing one *value* is cheap, because the product is never expanded.

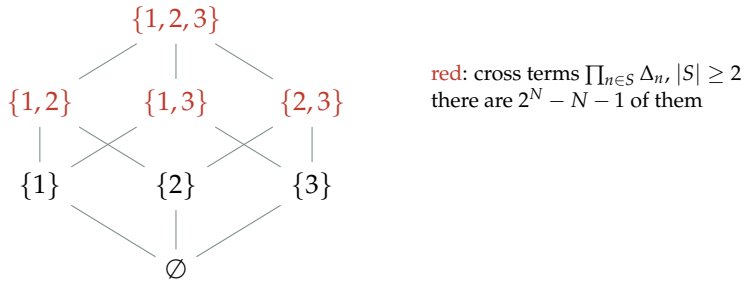


Figure 8.2: The subset lattice for $N = 3$ terms. Each node with at least two elements is one cross term of the interaction.

Theorem 8.6 Cost of the interaction (P1-T3) [proved]

Let the datum be a sum of M exponentials with polynomial prefactors of degree at most L, and let $d_n = \deg a_n$.

(a) One value of E_{int} costs

$$O\Big(M\big[N(L+1)^2 + (L+1)\textstyle\sum_n (d_n+1)\big]\Big);$$

for simple exponentials ($L = 0$) this is $O\big(M(N + \sum_n d_n)\big)$: linear in the number of terms.

(b) All aggregates by order, $E^{(k)} = \sum_{|S|=k}\prod_{n\in S}\Delta_n u_0$ for $2 \leq k \leq N$, cost $O(MN^2(L+1)^2)$ plus the same term in d_n.

(c) The full table of cross terms has $M(2^N - N - 1)$ entries and costs $\Theta\big(M\,2^N(L+1)^2\big)$.

Dim. P1-T3, Catalogue p. 35

Example 8.7 Counting operations for one mode

For one simple mode $e^{\zeta x}$ and N terms, compute the N numbers $z_n = e^{(t-t_0)a_n(\zeta)} - 1$, then the product $\prod_n (1+z_n)$ with $N-1$ multiplications, then subtract $1 + \sum_n z_n$. That is about $3N$ operations. Listing the $2^N - N - 1$ products $\prod_{n\in S} z_n$ one by one costs 2^N times more for large N: already for $N = 20$, about a million.

8.4 The flows are not determined without a class

Proposition 8.8 Non-uniqueness of single flows (P1-P1(a)–(d)) [proved]

On a bounded rectangle, without a function class:

(a) for the transport term $a_n = \xi$, the flow of a datum is any smooth extension of it: infinitely many choices;

(b) for the diffusion term $a_n = \xi^2$, there are nonzero heat solutions with zero datum (Tychonoff) [from cited source];

(c) so E_{int} is not a function of u alone; inside $\mathfrak{B}$ it is unique.

Dim. P1-P1, Catalogue p. 87

8.5 Stability: the interaction cannot be computed stably

Theorem 8.9 No continuous extension to Sobolev spaces (P1-P1(e)) [false]

Let $Q = \tau + c\xi - D\xi^2$ with $c, D > 0$ (advection–diffusion), terms $a_1 = -c\xi$ and $a_2 = D\xi^2$, and any base time t_0. For every m,

$$\sup \left\{ \frac{\|E_{\text{int}}\|_{L^2(\Omega)}}{\|u\|_{H^m(\Omega)}} : u \in \text{EP},\ Qu = 0,\ u \neq 0 \right\} = \infty.$$

So $u \mapsto E_{\text{int}}$ does not extend continuously from exponential-type data to H^m, although exponential-type data are dense. The same holds for every evolution with at least two non-proportional terms; for proportional terms the interaction is continuous (OPA-P3) [proved].

Dim. P1-P1e, Catalogue p. 37

Intuition

On a bounded interval, a real exponential $e^{\zeta x}$ with large ζ is a legitimate datum. The diffusion flow alone amplifies it by $e^{D\zeta^2 t}$. The true solution is only changed by the factor $e^{(D\zeta^2 - c\zeta)t}$ times the datum norm. The interaction term must cancel the gap, and the gap grows faster than any Sobolev norm of the datum.

This instability has three consequences for any algorithm that wants to be stable.

Theorem 8.10 No regularisation removes the interaction (P2-T1, P2-T2, P2-T4) [proved]

(a) **Any penalty leaves the same residual.** For any regulariser (an ℓ^1 sparsity penalty, ℓ^2, total variation, ...) the residual of the best fit converges to the distance of u from the closed span of the per-term pieces.

(b) **Inside an evolution factor that residual is large.** If no single term exhausts the symbol, the part of u outside the span of the per-term kernels satisfies

$$\|E^{\perp}(u)\|_{H^m} \;\geq\; \frac{\|g(\partial_x)u\|_{H^{m-k}}}{C}, \qquad g = \prod_n (a - a_n).$$

Only a finite-dimensional space of exponential-polynomials has a per-term representation. For advection–diffusion it is

$$\operatorname{span}\{1,\ x - ct,\ (x - ct)^2 + 2Dt\}.$$

(c) **No uniformly Lipschitz regularisation exists.** Any family of Lipschitz maps converging pointwise to an unbounded map has Lipschitz constants tending to infinity. Every exact selection of components has Lipschitz constant at least $\gamma_N^{-1/2}$.

Dim. P2-T2, Catalogue p. 38

The one continuous split found

A *balance* decomposition of u writes each term's contribution as the time integral of that term applied to the true solution:

$$u^{(i)}(t) = \int_{t_0}^{t} a_i(\partial_x)\, u(s)\, \mathrm{d}s.$$

It is exact ($u - u_0 = \sum_i u^{(i)}$) and continuous in H^m (P3-T6) [proof written, not audited]. But it needs an external principle to decide that this is "the" contribution of term i: it uses the full solution, not the flow of the term alone.

8.6 Time-dependent perturbation theory

The product formula needs commuting terms. With non-commuting terms, or with boundary conditions, there is no closed formula, but E_{int} is still defined by (8.1). Perturbation theory expands the flow in powers of the interaction. Split $A = A_1 + B$, where A_1 is the principal term and B the rest. The Dyson–Phillips series is

$$\begin{aligned} S(t) &= \sum_{k \geq 0} S^{(k)}(t), \\ S^{(0)}(t) &= e^{tA_1}, \\ S^{(k+1)}(t) &= \int_0^t e^{(t-s)A_1}\, B\, S^{(k)}(s)\, \mathrm{d}s. \end{aligned} \tag{8.3}$$

$S^{(k)}$ collects the histories in which B acts k times, at intermediate times, with the flow of A_1 in between. This is the interaction picture of quantum mechanics [10].

Theorem 8.11 Dyson–Phillips converges for every time (P3-T1) [proved]

Let $A_1 = a(\partial_x)$ with symbol of degree q, and $B = \sum_{k \le r} b_k(x)\partial_x^k$ with bounded coefficients and $r < q$. The symbol condition

$$\operatorname{Re} a(i\kappa) \le \omega - \delta|\kappa|^q \qquad (\delta > 0)$$

holds if and only if q is even and the leading coefficient has negative real part (a dissipative operator of even order). Under it, the Dyson–Phillips series converges in operator norm, uniformly on every bounded time interval, and its sum is the semigroup generated by $A_1 + B$. The same holds with Dirichlet conditions on an interval.

Dim. P3-T1, Catalogue p. 40

So E_{int} is well defined and can be approximated by truncating the series even without a closed formula.

Proposition 8.12 Interaction created by a boundary (P1-P6, P3-T3, OPA-T2, OPF5A-T5) [proved]

On an interval with Dirichlet conditions, advection and diffusion no longer commute: the boundary creates an $E_{\text{int}}^{\text{mult}}$ already at first order (P1-P6). For small times its size is governed by how the datum touches the boundary. If k is the order of contact (the first nonvanishing derivative of u_0 at the endpoints), then

$$\|E_{\text{int}}^{\text{mult}}(t)\| = \Theta\big(t^{k/2+3/4}\big) \qquad (t \to 0).$$

For compatible data the leading constant is explicit and proportional to

$$|c|\,\frac{\tilde{N}_k}{k!}\,D^{k/2-1/4}\sqrt{u_0^{(k)}(0)^2 + u_0^{(k)}(L)^2}.$$

Dim. OPF5A-T5, Catalogue p. 91

The smoother the datum is against the boundary, the faster the boundary interaction becomes negligible as $t \to 0$.

8.7 Non-commuting terms

If two terms do not commute, the flow of their sum is not the product of the flows. By the Baker–Campbell–Hausdorff formula,

$$e^{tA_1}e^{tA_2} = \exp\Big(t(A_1 + A_2) + \frac{t^2}{2}[A_1, A_2] + O(t^3)\Big).$$

The commutators generate a Lie algebra, and its type decides what can be done.

Theorem 8.13 Lie-algebra classification (P1-T4, P1-P5, P1-T5) [proved]

(a) **Finite and solvable.** An exact formula exists, with $d = \dim \mathfrak{g}$ entire scalar functions (Wei–Norman). Example: $\partial_x^2 + x$, and the Ornstein–Uhlenbeck operator.

(b) **Finite, not solvable** (for example $\mathfrak{sl}_2$). A closed form exists locally in time; it can fail globally.

(c) **Infinite adjoint orbit.** If X and all the iterated commutators $\mathrm{ad}_X^k Y$ are linearly independent (for example $X = x^a\partial_x$, $Y = \partial_x$ with $a \geq 3$), then no finite product of flows of the two terms, with times analytic in t and in the coupling ε and with the flows of Y switched off at $\varepsilon = 0$, reproduces $e^{t(X+\varepsilon Y)}$. Only series expansions exist.

Dim. P1-T5, Catalogue p. 42

Example 8.14 Which potentials allow Wei–Norman

For $i\partial_t = -\partial_x^2 + V(x)$, the operators ∂_x^2 and x^2 generate the finite Lie algebra $\mathfrak{sl}_2$: the harmonic oscillator has a closed-form evolution. With $V = x^3$, repeated commutators produce polynomials of ever higher degree and the algebra is infinite-dimensional. But the hypothesis of part (c) fails there, so whether a finite product of flows is impossible for ∂_x^2 and x^k with $k \geq 3$ is still [open] (P1-Q3). Part (c) applies to the transport with polynomial speed $u_t = (x^a + \varepsilon)u_x$, $a \geq 3$, which nevertheless has an explicit solution by characteristics: the theorem excludes only a factorisation into flows of the terms.

Other ways commutativity is lost

- **Repeated factors.** For $(\tau - a)^e$ the "multiplicity term" is a nilpotent shift. It commutes with the other terms, its flow is polynomial in time (secular growth), and $E_{\mathrm{int}}^{\mathrm{mult}} = 0$ (OPF-T2) [proof written, not audited].
- **Second-order evolutions** (wave, Klein–Gordon). The companion blocks do not commute, so the first-order calculus fails; a Gauss factorisation works mode by mode (II-T7, P1-P4) [proved].

Takeaway. For commuting terms the interaction is explicit and cheap to evaluate, but it can be neither removed nor computed stably in Sobolev spaces. That instability is not a defect of the formula: it is a property of the question. For non-commuting terms even the explicit form is lost, except when the terms generate a finite solvable Lie algebra.

CHAPTER 9

Nonlinear equations

In this chapter

Starting from	a scalar evolution in one space variable, $u_t = \sum_n T_n(u)$, each term T_n linear or polynomial in the jet $(u, u_x, u_{xx}, \dots)$; a smooth datum u_0; the exact solution is known. Benchmarks: viscous Burgers, KdV–Burgers, Fisher–KPP; KdV and NLS as integrable controls.
Why	for a nonlinear equation the single-term flows do not commute and superposition fails. "Component per term" needs a new definition, and identification needs a new frame (a dictionary). We give both and compute them where everything is explicit.
Role	fourth link of the chain. It reuses the earlier chapters through the jet expansion and the annihilator idea. Its results are weaker and relative, but sharp on the benchmark models.

9.1 The interaction term for nonlinear flows

Let S_n^t be the flow of $u_t = T_n(u)$ alone, which is now a nonlinear map. Definition (8.1) still makes sense:

$$E_{\text{int}}(t) = u(t) - u_0 - \sum_n \big(S_n^t u_0 - u_0\big).$$

There is no product formula, because the flows do not commute. But the Taylor expansion in time is explicit.

Theorem 9.1 Jet of the nonlinear interaction (S4T-T1) [proved]

For smooth data,

$$E_{\text{int}}(t) = \frac{t^2}{2} \sum_{n \neq m} T_n'[u_0]\big(T_m(u_0)\big) + O(t^3),$$

where $T_n'[u_0](v)$ is the derivative (Fréchet derivative) of T_n at u_0 in the direction v.

Dim. S4T-T1, Catalogue p. 45

Intuition

To first order in t every flow moves u_0 by $t\,T_n(u_0)$, and these moves add exactly: there is no interaction at order t. At order t^2 the full flow sees each term react to the motion caused by the others. That reaction is the derivative of T_n in the direction of $T_m(u_0)$.

Example 9.2 Burgers

For $u_t = -uu_x + \nu u_{xx}$ take $T_1(u) = -uu_x$ and $T_2(u) = \nu u_{xx}$. The derivatives are

$$T_1'[u](v) = -(vu_x + uv_x), \qquad T_2'[u](v) = \nu v_{xx}.$$

Substituting the other term:

$$T_1'[u_0]\big(T_2(u_0)\big) = -\nu\left(u_0''u_0' + u_0u_0'''\right),$$
$$T_2'[u_0]\big(T_1(u_0)\big) = -\nu\left(3u_0'u_0'' + u_0u_0'''\right).$$

Half the sum, times t^2, gives

$$E_{\text{int}}(t) = -\nu\left(u_0u_0''' + 2u_0'u_0''\right)t^2 + O(t^3).$$

9.2 Before, at and after the shock

The flow of the single term $-uu_x$ (inviscid Burgers) forms shocks: along each characteristic u is constant and the characteristic moves with speed u, so faster parts overtake slower ones. For $u_0 = \sin x$ the characteristics are

$$x = \xi + t\sin\xi,$$

and since

$$\frac{\partial x}{\partial \xi} = 1 + t\cos\xi$$

first vanishes at $\xi = \pi$, the breaking time is $t^* = 1$ (Figure 9.1).

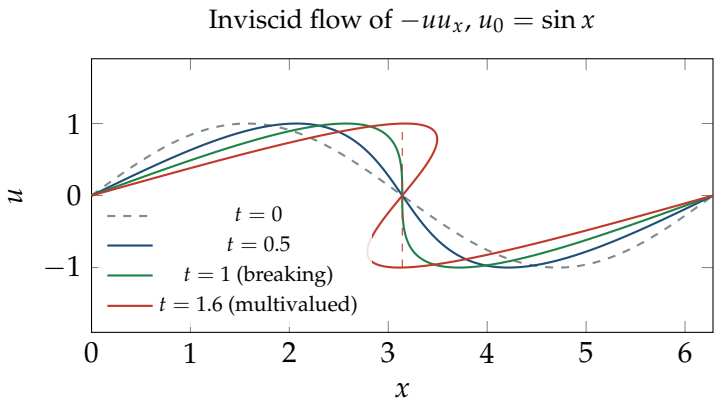


Figure 9.1: At $t = 1$ the profile has a vertical tangent and the gradient blows up. Beyond that the curve folds; the physical (entropy) solution replaces the fold by a jump at $x = \pi$. This breaking is what makes the interaction term discontinuous after the shock.

The interaction changes qualitatively with time, and each regime needs different tools.

9.2.1 Before the shock: a closed form

Theorem 9.3 Closed form before the shock (S4T-T2, S4T-P3) [proved]

For Burgers with $t < t^*$, the interaction term has a closed form at first order in the viscosity, with remainder

$$O(\nu^2 \varepsilon^2 t^3),$$

where ε measures the size of the datum. The closed form blows up at the breaking time t^*.

Dim. S4T-T2, Catalogue p. 46

The tool is the Cole–Hopf substitution

$$u = -2\nu \frac{\varphi_x}{\varphi},$$

which turns Burgers into the heat equation $\varphi_t = \nu \varphi_{xx}$ [13]. It plays the role of an "exact linearisation". For KdV and NLS the inverse scattering transform plays the same role.

The analyticity radius of the Burgers solution with datum U_0 is at least $2/\mathrm{osc}(U_0)$, and this bound is optimal (F4L-P3) [proved].

9.2.2 After the shock: sizes in L^1

The flow of $-uu_x$ alone forms a shock, so E_{int} has jumps whose size does not depend on ν (S4T-T3) [proved mod. [F1]]. This is forced by the definition. A function with a jump is not in H^1, so Sobolev norms are no longer the right measure. Pointwise values are unreliable, but the *measure* of the discrepancy is well behaved. The meaningful sizes are in L^1.

Theorem 9.4 Interaction after the shock (OPA-T1) [proved mod. Kruzkov]

At a fixed time $t > t^*$, under a nondegeneracy condition (H_t) that holds for example for $u_0 = \sin x$,

$$\|E_{\mathrm{int}}(t)\|_{L^1} = \Theta(\nu) \qquad (\nu \to 0).$$

Each shock of strength ρ contributes

$$4\nu \log\left(2\cosh(\rho/2)\right)$$

to the leading term.

Dim. OPA-T1, Catalogue p. 48

Theorem 9.5 Uniform rate across the shock time (OPF5A-T1, OPF5A-T2) [proved mod. Kruzkov]

Uniformly in time, near the shock time,

$$\|E_{\mathrm{int}}(t)\|_{L^1} \approx 2\nu \log \frac{1}{\max(|t - t^*|, \nu)} + O(\nu),$$

so that

$$\sup_{t \le T} \|E_{\text{int}}(t)\|_{L^1} = \nu |\log \nu| + O(\nu),$$

with leading coefficient exactly 1. The constant at the shock time is explicit (it involves the Pearcey constant 1.188654…). For data with several shock births the uniform rate is [conditional] on a localisation lemma.

Dim. OPF5A-T2, Catalogue p. 50

Proposition 9.6 The splitting error is larger (OPF5D-T6) [proved]

The error of the splitting scheme "advection, then diffusion" is

$$\|S(t)u_0 - S_2^t S_1^t u_0\|_{L^1} = \Theta(\sqrt{\nu}),$$

larger than the interaction term itself. Treating the terms one after the other costs more than ignoring their interplay.

Dim. OPF5D-T6, Catalogue p. 100

Regime	What happens	Tools and results
before, $t < t^*$	smooth; characteristics do not cross	jet (Thm. 9.1), closed form (Thm. 9.2.1), Cole–Hopf
at breaking, $t = t^*$	the gradient blows up	smoothness estimates fail; the closed form blows up
after, $t > t^*$	the single-term flow has a jump	L^1: $\Theta(\nu)$, uniform $\nu\|\log \nu\|$, splitting $\Theta(\sqrt{\nu})$

Table 9.1: The three regimes of the Burgers interaction term.

9.3 Identification relative to a dictionary

Nonlinear terms cannot be handled by the ideal $\mathrm{Ann}(u)$: there is no ring of operators. Instead one fixes a finite list of candidates.

Dictionary setting

- A *dictionary* $\mathcal{D}$ is a finite set of monomials in the jet $(u, u_x, u_{xx}, \dots)$ of degree at most d and order at most k, for example $\{u,\ u_x,\ uu_x,\ u_{xx},\ u_{xxx}\}$. $V = \operatorname{span} \mathcal{D}$.
- $I(u)$ is the set of all differential polynomial relations that u satisfies.
- $[F]$ is the differential ideal generated by the equation F: F, its derivatives $\partial_x F, \partial_t F, \dots$, and their combinations.
- $\mathcal{J}_u$ is a jet-level obstruction defined from u, and ψ maps the dictionary into the jet space.

Theorem 9.7 Dictionary criterion (S4ID-T1) [proved]

For smooth u on any Ω there is an exact sequence

$$0 \to [F] \cap V \to I(u) \cap V \to \mathcal{J}_u \cap \psi(V) \to 0.$$

In particular

$$I(u) \cap V = [F] \cap V \iff \mathcal{J}_u \cap \psi(V) = 0.$$

Dim. S4ID-T1, Catalogue p. 52

The correct statement is therefore "the relations are exactly F *modulo its differential consequences*". The literal statement $I(u) \cap V = \operatorname{span}\{F\}$ is false in complete dictionaries, because they contain $\partial_x F$ (S4ID-P1) [proved].

Theorem 9.8 Burgers with N shocks (S4ID-T2, OP-T7) [proved]

For the N-shock solutions of Burgers, u identifies its equation in the evolutionary dictionary with d unknown terms if and only if $d \le N$.

Dim. S4ID-T2, Catalogue p. 53

Example 9.9 One shock, two unknowns

The travelling front $u = -\tanh(x/(2\nu))$ is a 1-shock solution. It satisfies Burgers, but also the ODE that every travelling wave satisfies. With $d = 2$ unknown coefficients (in front of uu_x and u_{xx}) and $N = 1$, the theorem says u does not identify the pair. It fixes only their ratio.

Theorem 9.10 KdV N-solitons (S4ID-T3, OPF5C-T2) [proved mod. Novikov 1974]

For the N-soliton solution of KdV, the annihilator in the jet variables is principal, generated by one polynomial Π_N of total degree

$$\deg \Pi_N = (N+2)\, 2^{N-1}.$$

For generic $N = 2$ this degree is 8. The solution identifies KdV in a dictionary of degree d if and only if $d < (N+2)2^{N-1}$.

Dim. OPF5C-T2, Catalogue p. 54

Proposition 9.11 A closed form does not identify (S4ID-T4) [proved]

If u has a closed form, it lives on a smaller variety and satisfies extra relations: it does not identify its equation in a complete dictionary.

Dim. S4ID-T4, Catalogue p. 103

This is the nonlinear analogue of "exponential-polynomials never identify".

9.4 Integrable models and scattering

- **Hirota N-solitons.** The interaction is pairwise and localised: it is determined by pairs of solitons, integrates to 0, and costs $O(N^3)$ (INT-P5, INT-P6) [proved].
- **Scattering identification.** For reflectionless KdV, Novikov's centraliser characterises the potential (INT-T1, OPF5D) [proved]. For NLS with multiplicity, identification by the scattering data holds; without multiplicity it is [false]. The Lax and zero-curvature relations hold for all n (OP-P17, OP-P18) [proved].
- **KdV–Burgers.** The jet is proved. Before the shock the dispersive correction has remainder $O\big((\nu + |\delta|)^2\big)$ (OPR2-T5) [proved mod. local well-posedness].
- **Fisher–KPP.** Only a jet-level statement with an estimate that is not uniform in time (S4T-P5) [proved]. No sharp analogue of the Burgers asymptotics is claimed.

Open in the nonlinear part

For analytic KdV data with nonzero reflection, is there an autonomous ODE in x? Three obstructions are proved, among them the cubic relation of Airy [open] (INT-Q1). The exact minimal syzygy degrees $\delta_N(K)$ are known in nine computer-assisted cases; their general formulas are conjectures [open] (OPF5C-Q1).

9.5 Linear versus nonlinear

	Linear (Chapters 3–8)	Nonlinear (this chapter)
Candidate equations	the ring $\mathcal{R} = \mathbb{C}[\tau, \xi]$	a finite dictionary $V = \operatorname{span} \mathcal{D}$
Relations of u	the ideal $\operatorname{Ann}(u)$	the differential ideal $I(u)$
What P generates	the principal ideal (P)	the differential ideal $[F]$
Identification	$\operatorname{Ann}(u) = (P)$	$I(u) \cap V = [F] \cap V$
What escapes	the point spectrum $V(J_u)$	the obstruction $\mathcal{J}_u \cap \psi(V)$
Solutions that never identify	exponential-polynomials	closed forms
Interaction	product formula, $O(N)$	t^2 jet; $\Theta(\nu)$, $\nu\lvert\log\nu\rvert$ after shocks

Table 9.2: The linear and nonlinear problems side by side.

Takeaway. Nonlinearity changes the objects (single-term flows with shocks, dictionaries instead of rings) but not the question. Where the formula is closed (before the shock, solitons) the answer is explicit. Across a shock the interaction does not vanish with the viscosity: it shrinks like $\nu\lvert\log\nu\rvert$.

CHAPTER 10

The algorithm

In this chapter	
Starting from	an input in one of the data models M1–M3 (Chapter 5) and, for each step, the hypotheses of the theorem that justifies it.
Why	to turn the theorems into one procedure with explicit inputs, outputs and guarantees.
Role	operational summary of Chapters 4–9. Table 10.1 lists the steps, Figure 10.1 gives the control flow, and two worked examples make every step concrete.

10.1 The pipeline

Step	Task	Result relied upon	Output
0	Input	data models M1–M3	exact u
1	Annihilator	Gram matrix of a dictionary	a candidate P
2	Factor	unique factorisation (F1)	factors P_i, multiplicities e_i
3	EP test	Lemma 3.3, Thm. 4.2, Ch. 5	a verdict per factor
4	Reduce	Thms. 4.3 and 4.3	G_u, J_u, certificate
5	Decompose	Thm. 6.3, Eq. (6.1), Thm. 7.3	components u_j, ambiguity $\delta(P)$, check γ_N
6	Terms	Thms. 8.2, 8.3, 8.7	log-symbol, E_{int}
7	Nonlinear branch	Thms. 9.3, 9.1	relative identification, flow-based E_{int}

Table 10.1: The steps of the algorithm: what each step does, which result justifies it, and what it returns.

10.2 The steps in detail

Step 1: obtain an annihilator

Let $\Phi(u) = (T_1 u, \dots, T_M u)^\top$ be the vector of M candidate terms applied to u, and

$$\mathbb{G}(u) = \int_\Omega \Phi\, \Phi^*\, \mathrm{d}\mu$$

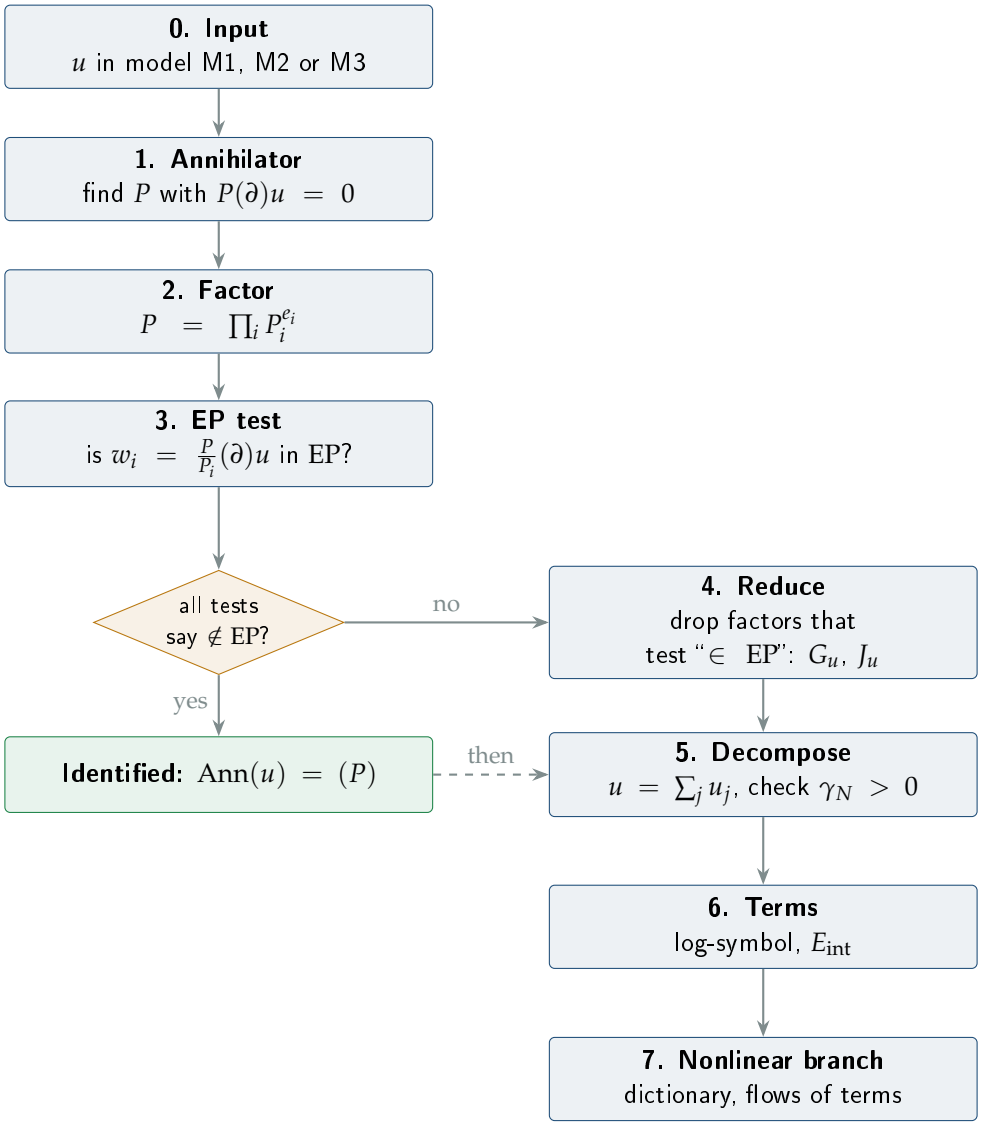


Figure 10.1: The algorithm as a flowchart. The left column certifies what the operator is. If every factor test says "not an exponential-polynomial", the operator is identified. Otherwise Step 4 reduces it to G_u. The right column extracts structure from u once the operator is known.

its Gram matrix (positive semidefinite). A relation $\lambda \cdot \Phi = 0$ holds if and only if $\lambda \in \ker \mathbb{G}(u)$. The computation is exact (rational or algebraic numbers, Gaussian elimination): no learning, no statistics. If $\ker \mathbb{G}$ has dimension larger than one, u satisfies several relations, which is itself information about G_u and J_u.

Step 2: factor

Factor $P(\tau, \xi)$ into irreducibles with multiplicities. This is a standard computer-algebra operation over the algebraic numbers.

Step 3: exponential-polynomial test

For each i compute $w_i = (P/P_i)(\partial)u$ and decide whether $\dim \mathcal{R}w_i < \infty$. How depends on the data model:

- M1: a syntactic check (is the closed form a finite sum of polynomials times exponentials?);
- M3: a holonomic computation, decidable modulo (AC) (Theorem 5.5);
- M2: undecidable from the jet alone (Theorem 5.3); with a bound B, compare annihilators up to degree $B + \deg P - 1$ (Theorem 5.4).

Step 4: reduce to G_u

Drop from P each factor whose test returned "EP", with the multiplicity bookkeeping of Theorem 4.3: at most $\sum_i e_i$ tests in total. The output is G_u; the dropped part is J_u, whose finite zero set lists the exponentials contained in u. The certificate reads either "$\mathrm{Ann}(u) = (P)$" or "u identifies only G_u and contains these exponentials".

Step 5: decompose

For evolution factors $\tau - a_j(\xi)$ apply the Lagrange formula (6.1). For general coprime factors use the construction of Theorem 6.3; the components are determined up to the $\delta(P)$-dimensional exponential-polynomial space, fixed by a normalisation. Before using the components in H^m, check $\gamma_N > 0$ (Theorem 7.3).

Step 6: terms and interaction

Inside an evolution factor with symbol $a = \sum_n a_n$, the terms are read from the log-symbol, which is linear in time:

$$\log \hat{u}(t,\kappa) = \log \hat{u}_0(\kappa) + t \sum_n a_n(i\kappa).$$

Then evaluate E_{int} by the product formula (8.2) at cost $O(N)$ per mode. For non-commuting terms, classify the generated Lie algebra (Theorem 8.7).

Step 7: nonlinear case

Replace Step 3 by the dictionary criterion (Theorem 9.3) and Step 6 by the flow-based interaction term and its jet (Theorem 9.1).

10.3 Worked example 1: the wave equation

We follow Steps 1–5 on the wave equation with $c = 1$ and the solution

$$u(t,x) = \underbrace{\frac{1}{1+(x+t)^2}}_{f(x+t)} + \underbrace{e^{-(x-t)^2}}_{g(x-t)}.$$

Example 10.1 Step 1: annihilator

Take the dictionary $\{u_{tt}, u_{xx}, u_{tx}, u_t, u_x, u\}$. The kernel of the Gram matrix is the line spanned by $u_{tt} - u_{xx} = 0$. Output:

$$P = \tau^2 - \xi^2.$$

Example 10.2 Step 2: factor

$$P = Q_1 Q_2, \qquad Q_1 = \tau - \xi, \qquad Q_2 = \tau + \xi.$$

Two irreducible, simple factors.

Example 10.3 Step 3: EP tests

Remove Q_1. We test $w_1 = (\partial_t + \partial_x)u$. The two parts behave differently:

$$(\partial_t + \partial_x) f(x+t) = 2f'(x+t), \qquad (\partial_t + \partial_x) g(x-t) = 0.$$

With $f(s) = 1/(1+s^2)$:

$$w_1 = -\frac{4(x+t)}{\left(1+(x+t)^2\right)^2}.$$

This is rational, not a finite sum of polynomials times exponentials: $w_1 \notin \mathrm{EP}$.

Remove Q_2. Symmetrically, with $g(s) = e^{-s^2}$:

$$w_2 = (\partial_t - \partial_x)u = -2g'(x-t) = 4(x-t)\, e^{-(x-t)^2}.$$

A Gaussian times a polynomial is not an exponential-polynomial: $w_2 \notin \mathrm{EP}$.

Example 10.4 Step 4: reduce

Both tests say "not EP", so nothing is dropped. The point spectrum is empty, and the wave equation is identified:

$$\mathrm{Ann}(u) = (\tau^2 - \xi^2).$$

Example 10.5 Step 5: decompose

$(Q_1, Q_2) = (\tau, \xi)$, so $\delta(P) = 1$. The components are $u_1 = f(x+t)$ and $u_2 = g(x-t)$, determined up to a constant moved from one to the other. Whether the components of every solution stay in H^m on a given region is decided by Theorem 7.3; on a square it holds (Example after Theorem 7.3).

10.4 Worked example 2: advection–diffusion

Now take the solution of Example 8.2,

$$u = g(x - ct;\ \sigma^2 + 2Dt), \qquad c = 2,\ D = \tfrac{1}{2},\ \sigma^2 = 1.$$

Example 10.6 Steps 1–4

The Gram matrix of $\{u_t, u_x, u_{xx}, u\}$ has kernel spanned by $u_t + 2u_x - \frac{1}{2}u_{xx}$, so

$$P = \tau + 2\xi - \tfrac{1}{2}\xi^2.$$

It is irreducible (degree one in τ). The test is on u itself: a Gaussian is not an exponential-polynomial, so $\mathrm{Ann}(u) = (P)$ and the equation is identified.

Example 10.7 Step 5

There is a single factor: no decomposition is needed.

Example 10.8 Step 6: terms and interaction

The log-symbol is

$$\log \hat{u}(t,\kappa) = \log \hat{u}_0(\kappa) + t\big(-2i\kappa - \tfrac{1}{2}\kappa^2\big).$$

Its two coefficients are the two terms: advection with speed 2 and diffusion with constant $\frac{1}{2}$. The interaction term follows from (8.2), at cost $O(2)$ per Fourier mode:

$$\widehat{E_{\mathrm{int}}}(t,\kappa) = \big(e^{-2i\kappa t} - 1\big)\big(e^{-\kappa^2 t/2} - 1\big)\,\hat{u}_0(\kappa).$$

Figure 8.1 plots it.

10.5 Cost and feasibility

Component	Cost or feasibility	Source
annihilator from a dictionary	Gaussian elimination, $O(M^3)$	Step 1
factorisation	standard computer algebra	Step 2
EP test, model M1	syntactic	Step 3
EP test, model M3	holonomic, decidable mod (AC)	Thm. 5.5
EP test, black box	impossible without a bound	Thm. 5.3
EP test with bound B	linear algebra in degree $B + \deg P - 1$	Thm. 5.4
components (evolution)	one rational multiplier per factor	Eq. (6.1)
E_{int}: value / by order / table	$O(N)$ / $O(N^2)$ / $\Theta(2^N)$	Thm. 8.3

Table 10.2: Cost and feasibility of the components of the algorithm.

Takeaway. Every step of the algorithm is a theorem with explicit hypotheses. The algorithm is exact where the theorems are positive (Steps 2, 4, 5 and 6 on exponential-type data), and it reports honestly where they are negative (Step 3 on finite data, Step 6 in Sobolev spaces).

CHAPTER 11

Limits, open problems and status

In this chapter	
Starting from	no new assumptions.
Why	to state what is impossible, what is open, which statements turned out false, and how reliable each result is.
Role	delimits every claim of this edition.

11.1 What is impossible

Each item below is a proved negative theorem, not a gap.

- a finite certificate of identification, and identification from a black box (Theorems 5.3 and 5.3);
- identification by restricting variable-coefficient operators to one solution (Proposition 5.6);
- decomposition of a solution by additive terms (Proposition 6.1.1);
- factor decomposition on regions with holes, for some P (Proposition 6.4);
- components in H^m in general: one derivative can be lost (Counterexample 7.1);
- a continuous or stably regularised interaction term in H^m (Theorems 8.5 and 8.5);
- a finite product of single-term flows reproducing non-commuting dynamics when the adjoint orbit is infinite (Theorem 8.7(c)).

11.2 Statements that turned out false

Statement	Counterexample or correction	Ledger
the spectrum of an expansion decides identification on a bounded domain	e^{x-t} on the unit square (Counterexample 4.4.2)	C3-C1
components keep the regularity of u	parallel transports on the square	C12-C1
$u \mapsto E_{\mathrm{int}}$ is continuous in H^m	real exponentials in advection–diffusion	P1-P1(e)
the companion matrix is normal iff $p = s^m - c$	$s^2 + s - 1$	R6-T3
splitting by terms lies in $\ker(L_1, L_2)$	$\frac{1}{2}x^2 + Dt$ for the heat equation	R2-T3 (corrected)
exact interaction impossible for infinite Lie algebras (as first stated)	$e^{-tT_2}e^{-tT_1}e^{t(T_1+T_2)}$	R5-Q5, P1-C1
NLS identification by scattering data without multiplicity	explicit counterexample	OPF5D

11.3 Open or conditional

- closedness of the sum of kernels for $N \geq 3$ differential factors [open] (C12-Q2); solved for real transports in L^2;
- density of the sum of kernels on non-convex regions [open];
- minimality of the bound B on every region [open] (OP-ALG-Q2);
- identification for analytic KdV data with nonzero reflection [open] (INT-Q1);
- the uniform shock rate with several shock births [conditional] (OPA-Q1);
- the exact values of the syzygy degrees $\delta_N(K)$ for general N [open] (OPF5C-Q1);
- no finite product of flows for ∂_x^2 and x^k, $k \geq 3$ (the hypothesis of Theorem 8.7(c) fails there) [open] (P1-Q3), and Theorem 8.7(c) without the assumption that the flows of Y vanish at zero coupling [open] (P1-Q7).

11.4 How reliable the results are

Proofs and audits. Every statement labelled [proved] has a complete proof that passed an independent review. The reviewers were AI agents with no access to the author's reasoning, not human referees. Statements labelled [proof written, not audited] have complete proofs that were not audited; none of them is on the critical path of the main theorems.

Formal verification. Fifteen theorems of the algebraic core are formalised in Lean 4 with Mathlib, without `sorry`: the interaction identity for commuting flows, the comaximal factor decomposition with explicit projector, the normality criterion for $m = 2$, and several irreducibility

results. The analytic parts (distributions, Sobolev spaces, semigroups) are not formalised, because the libraries are missing.

Citations. Many ingredients are classical: Malgrange–Ehrenpreis, the Nullstellensatz, Cole–Hopf, inverse scattering. The primary sources of the load-bearing citations (Malgrange, Fulton, Kružkov, Novikov) were checked; several others only on secondary sources. A result relied upon should be traced to its primary source.

Novelty. The contribution is the package: sharp characterisations, negative theorems and counterexamples, each with a verified status. Parts of the identification theory are folklore in commutative algebra (Oberst duality); parts of the finite-data theory are known in substance (Prony, Mourrain).

11.5 What this edition does not contain

- Phases 6–11: systems and matrix coefficients, variable coefficients and the analytic encoder, the universal dictionary, convergence, the extended algorithm. They are summarised in *Punto di partenza* and will enter the complete edition after Phase 11.
- The catalogue of about 60 equation types and the tropical reading of scaling limits.
- The Lean sources.

CHAPTER 12

Open problems and directions for improvement (Phases 0–5)

In this chapter	
Starting from	the final report of Phases 0–5 (26 October 2026), the registry of open items, the route registry (Gates 1 and 4a) and the note on a learning pipeline.
Why	Chapter 11 states in a few lines what is open. This chapter gives the full list for Phases 0–5, what is already proved about each item, the improvements that would strengthen the work, the routes that were scouted and not taken, and the limits of the project.
Role	a complete record. Items that were later closed, reduced or refuted in Phase 10 are marked; for those, the complete edition (Phases 0–10) prevails.

12.1 Still open or conditional at the end of Phase 5

Item	Question	Status and obstruction	Later
INT-Q1 (analytic data)	KdV, $r \not\equiv 0$, u analytic: is there an autonomous ODE in x?	open. New partial result OPF5C-T6: relations with non-singular origin force exponential tails; order ≤ 3 gives 0 or a 1-soliton (with decay on the whole jet). Residual obstruction: singular relations at the origin, exactly where the cubic relation of Airy sits	reduced to (R) in Phase 10
OPA-Q1 general	several shock births	conditional on the localisation lemma L-loc (statement written in the audit S-CHECK-20)	closed modulo [F1]
OPA-Q2 density	density of the class Morse+(SB)	a sketch (Sard with finitely many parameters)	density proved
OPR2-Q2′	wedges with degenerate directions	partial: the tangent model is closed and direct (OPF5B-P8); localisation to the polygon is missing	still open
OPR2-Q3	repeated socle, “no finite test”	partial: proved under (I) existence of an identifying solution and (II)	characterised in Phase 10

Item	Question	Status and obstruction	Later
OPF5C-Q1	$\delta_N(K)$ for general N and $N < K < 2N-2$	open; conjectures $\delta_N(2N-1) = N+2$, $\delta_N(2N-2) = N+4$, $\delta_N(2N) = N+1$	computer-assisted cases added
S4ID-P8, OPR2-P7	asymptotics with derivatives	conditional on [GT-u] (uniformity in x is not stated by Grunert–Teschl; track: Deift–Zhou 1994) and on [Hm]	[Hm] proved
INT-T1(B)	reflectionless Hirota N-solitons	conditional on [F4$^-$] (the primary source in this form was not found; verified for $N = 2$)	proved
non-convex regions	density of OPF5B-T1 outside convex sets	open	false for regions with holes

Other open questions recorded in the text.

- the closedness of the sum of kernels for $N \geq 3$ differential factors [open] (C12-Q2) (solved for real transports in L^2);
- minimality of the bound B on every region [open] (OP-ALG-Q2);
- no finite product of flows for ∂_x^2 and x^k, $k \geq 3$ [open] (P1-Q3), and Theorem 8.7(c) without the assumption that the flows of Y vanish at zero coupling [open] (P1-Q7);
- P1-Q6: no continuity of the interaction term for any operator with two non-proportional terms [open] (*false as written*, corrected in Phase 10).

Dependences that are modulo a source. [F1] Lax–Oleinik = Kružkov (the source was read; an autonomous proof is missing); [L-loc]; [GT-u]; [F4$^-$]; and Malgrange 1966 for C3-P1 with general data on convex sets (the jet case is proved without it).

12.2 Improving what is there

- **Citations whose numbers were not checked**: Krantz–Parks Thm 2.3.5 (OP-T1), Cohen GTM 138 for (AC), Hörmander SCV 1.4.4, Runge in Rudin. Of 112 sources, 34 had the theorem number checked and 75 the content (45 only in the metadata); three wrong citations were corrected (Humphreys Cor. B, Gel'fand–Dikii 1976, DLMF 9.7.10).
- **Audits**: several statements are [proof written, not audited] (not audited); TROP-5–8 are written but not audited; the Gaussian example of Chapter 8 and the Lagrange projector formula were derived for this edition and not literally in the ledger.
- **Algorithm**: the Gram-matrix procedure of Step 1 is one way to obtain a candidate annihilator and has to be confirmed as the intended one; a cost analysis of each step is only indicative; the nonlinear branch has no full worked example.

- **Fisher–KPP** is announced as a benchmark but only an estimate non-uniform in t is recorded (S4T-P5); a statement or a removal is needed.
- **Examples** suggested but not added: a worked example of the exact sequence of the dictionary criterion, and an example of a non-empty point spectrum with several factors.

12.3 Routes scouted and not taken (Phases 1–5)

Each candidate route was promoted, merged, parked or eliminated with a stated reason.

- **Promoted**: factor decomposition with the least common left multiple for variable coefficients (RT-1/R2b); the additive log-symbol for first-order evolutions (RT-2); the Wei–Norman and Dyson interaction analysis (RT-3); $\ker \mathbb{G} = \mathrm{Ann}(u) \cap \mathrm{span}$ and Prony $\Leftrightarrow$ Waring (RT-4); complete invariants (RT-6); the weighted Fock orthogonal basis (RT-R3a; extension to Airy open).
- **Parked**: the symbol-tensor route RT-5 (the tensor keeps only the principal symbol; kept as a geometric invariant); RT-S1e, RT-S1f (no constant-coefficient theorem; only sectorial components); RT-R1b (needs a sign hypothesis on the datum); Koopman operators ("dynamic mode decomposition is regression"; to reopen with the normal form: eigenfunctions carry the nonlinear term); Painlevé, NLSPEC, HAANTJES (triggers in the nonlinear verdicts).
- **Eliminated**: RT-S1g for the core (it gives no information for elliptic or parabolic operators), kept for the hyperbolic part.
- **Scouted in Phases 3–4**: the tropical estimate of E_{int} from the symbol alone holds only on dissipative faces (TROP-3′, TROP-8); on conservative faces (Klein–Gordon, transport) no uniform estimate exists (TROP-4).

12.4 The statistical wrapper that was proposed

A learning pipeline was drafted around the exact analytic core and *excluded* from the project by decision of the user (no machine learning). It is recorded as a direction: (1) an interpolation step returns a weighted family of smooth functions $(\hat{u}_s, \pi_s)$; (2) each is passed to the exact algorithm, which returns its factors and components; (3) the weights give a distribution over factors ("transport 0.7 ± 0.1, diffusion 0.3 ± 0.1"); (4) the distribution sets the parameters of a clustering, so that each group contains one factor; (5) repeat inside each group, with two measures of "tangle": the Friedrichs cosine between factors and E_{int} between terms. The exact parts exist; the interpolation, aggregation and clustering are a proposal, and the note asked whether the basis functions of the first step must be chosen in advance.

12.5 What this edition does not contain

The catalogue of about 60 equation types and the tropical reading of scaling limits; the Lean sources; a human review (the reviewers were AI agents); Phases 6–10, which are in the complete edition.

APPENDIX A

Glossary of classical tools

Each entry states a classical tool in one sentence and points to where this edition uses it.

Name	Statement in one sentence	Used in
Unique factorisation	Polynomials in two variables factor uniquely into irreducibles, up to constants.	Ch. 3
Bézout's theorem	Two curves without a common component meet in at most the product of their degrees points [2].	Ch. 3, 5
Hilbert's Nullstellensatz	A polynomial vanishing on the zero set of an ideal has a power in the ideal.	Ch. 4
Chinese remainder theorem	Coprime moduli split a quotient ring into a product.	Ch. 6
Malgrange–Ehrenpreis	Every nonzero constant-coefficient operator has a fundamental solution, and is onto on smooth functions on a convex set [3, 4].	Ch. 6, 5
Oberst duality	Solution spaces of constant-coefficient systems correspond to modules over the operator ring [6].	Ch. 4
Poincaré lemma	On a simply connected region every closed 1-form is exact.	Ch. 6
Friedrichs angle	$M + N$ is closed iff the cosine of the angle between M and N, after removing $M \cap N$, is smaller than 1.	Ch. 7
Miyadera–Voigt	Small relatively bounded perturbations of a generator generate a semigroup [9].	Ch. 8
Baker–Campbell–Hausdorff	$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\cdots}$.	Ch. 8
Wei–Norman	Linear time-dependent systems on a solvable Lie algebra reduce to ODEs for scalar coefficients [11].	Ch. 8
Tychonoff	The heat equation has nonzero smooth solutions with zero initial datum [16].	Ch. 8
Cole–Hopf	$u = -2\nu\varphi_x/\varphi$ maps Burgers to the heat equation [13].	Ch. 9
Kružkov	Entropy solutions of scalar conservation laws are unique and L^1-stable [14].	Ch. 9
Inverse scattering	KdV and NLS are linearised by a nonlinear Fourier transform with trivial time evolution.	Ch. 9
Novikov	The stationary KdV hierarchy characterises finite-gap and soliton potentials [15].	Ch. 9

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